

Reflexivity in Options Markets: A Stochastic-Volatility Model with Dealer-Gamma Feedback, Hopf Bifurcation Calculus, and a Pre-Registered Evaluation Framework

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Abstract

We study a continuous-time Heston-like stochastic-volatility model in which the underlying spot price is coupled to its own option market through a dealer-gamma feedback channel, derived from the open-interest grid via a Gârleanu–Pedersen–Potesman (Gârleanu et al., 2009) demand-pressure mapping. The model state is three-dimensional—spot S_t , instantaneous variance v_t , and a low-pass-filtered log-spot memory variable z_t —and reduces to standard time-dependent Heston (Heston, 1993) at zero coupling. Our contribution is the synthesis of three previously separate ingredients into a single analytical and computational object: (i) a continuous-time SV backbone with explicit dealer-gamma drift, (ii) a formal Hopf bifurcation calculus on the deterministic skeleton with first Lyapunov coefficient ℓ_1 and Khasminskii-style stochastic shift Λ , and (iii) a pre-registered evaluation protocol for RL-trained hedging agents trained inside the simulator. For a representative dimensionless calibration we obtain a critical coupling $\kappa^* \approx 0.8964$, angular frequency $\omega^* \approx 0.5724$ rad/yr, and first Lyapunov coefficient $\ell_1 \approx -0.025 < 0$, so the bifurcation is supercritical: an attracting limit cycle in volatility is born for $\kappa > \kappa^*$. The 2D reduction (without z_t) has an upper-triangular Jacobian and admits only a saddle-node onset, recovering the discrete-time recursion of Dai (2025); the 3D extension is what unlocks the Hopf. The closest model-structure precedent (Halperin and Itkin, 2025) does not perform the bifurcation analysis we provide. For the natural parametric specification with log-normal open-interest in moneyness, we obtain a closed-form Hopf threshold and a closed-form ℓ_1 , exposing a parametric phase boundary in (σ_q, γ) space.

1 Introduction

Real options markets exhibit endogenous volatility cycles that pure stochastic-volatility (SV) models cannot reproduce. The empirical anchor is Hardiman et al. (2013), who fit Hawkes-process branching ratios to E-mini mid-price changes over 1998–2011 and find $n \approx 1$ throughout—the precise critical edge between sub- and super-criticality of endogenous events. Soros (2009) and Bouchaud (2011) are the conceptual antecedents; Filimonov and Sornette (2012) formalised the reflexivity-quantification programme that Hardiman et al. (2013) executed. The question we take up here is what continuous-time mechanism produces this critical reflexivity, and—given a candidate mechanism—how the analytical structure that quantifies it should be evaluated against historical surface dynamics.

We contribute the following.

- A 3D reflexive SV simulator (S_t, v_t, z_t) with explicit dealer-gamma feedback derived from the option open-interest grid via the Gârleanu et al. (2009) demand-pressure mapping. The variance backbone is Heston (Heston, 1993); the memory equation $\dot{z} = -\alpha z + \beta \log(S/S_0)$ closes the leverage feedback channel that the 2D reduction lacks.
- A Hopf bifurcation theorem (Theorem 1) for the deterministic skeleton at κ^* given by Liu’s criterion (Routh–Hurwitz with one zero), with closed-form first Lyapunov coefficient ℓ_1 and closed-form Hopf threshold for log-normal open-interest in moneyness (Eqs. 14–15).
- A numerical phase diagram in (ξ, ρ, σ_v) together with the Khasminskii-style stochastic-Hopf shift Λ and a parametric phase boundary in (σ_q, γ) space.
- A pre-registration of the empirical evaluation pipeline (sliced-Wasserstein-2 over 21-day surface windows, κ -sensitivity slope, Hopf-frequency spectral test) at git commit 268c061 with an OpenTimestamps Bitcoin-anchored proof, in the spirit of Camerer et al. (2018) adapted for computational finance.

Closest precedents. Halperin and Itkin (2025) (Marketron) share the model-structural axis—continuous-time SDE, memory channel, options-aware—but do not perform the bifurcation analysis. Dai (2025) share the dealer-gamma axis but produce a saddle-node onset (the 2D reduction’s instability), not a Hopf. He et al. (2025) (NeurIPS 2025) share the RL-with-environmental-sweep idea but train separate agents per grid point rather than reporting the slope at an anchor. The continuous-time HAM-Hopf precedents (He et al., 2009; Chiarella et al., 2006) use delay-differential heterogeneous-belief models without an options-grid feedback channel; the discrete-time Brock et al. (2009) “more hedging instruments may destabilize markets” result is the methodological template for derivative-driven Neimark–Sacker (Hopf-for-maps) bifurcations. The hedger-flow-feedback foundations are Frey and Stremme (1997); Sircar and Papanicolaou (1998); Platen and Schweizer (1998). Sections 3, 5 and 6 make these contrasts precise.

Roadmap. Section 2 states the model. Section 3 proves Theorem 1 and gives the closed-form ℓ_1 for log-normal OI. Section 4 reports the numerical phase diagram. Section 5 describes the pre-registered evaluation framework. Section 6 reports the mechanism decomposition against Marketron. Section 7 closes with the empirical-calibration roadmap. Code at <https://github.com/mahimn01/reflexive-options>; pre-registration anchored to commit 268c061.

2 The reflexive model

Let S_t be the underlying spot price, v_t the instantaneous variance, and z_t a memory variable encoding low-pass-filtered price history. The reflexive system is

$$\frac{dS_t}{S_t} = (\mu + \kappa G(S_t, z_t, v_t)) dt + \sigma(S_t, v_t) dW_t^S, \quad (1)$$

$$dv_t = (\kappa_v(\theta_v - v_t) + \gamma z_t) dt + \xi \sqrt{v_t} dW_t^v, \quad (2)$$

$$dz_t = (-\alpha z_t + \beta \log(S_t/S_0)) dt, \quad (3)$$

$$d\langle W^S, W^v \rangle_t = \rho dt, \quad (4)$$

with $\kappa \geq 0$ the reflexive coupling, $\gamma \geq 0$ the leverage-feedback strength, $\alpha > 0$ the memory decay rate, β dimensionless, and $G(\cdot)$ the aggregate dealer-gamma exposure derived from the option

open-interest grid:

$$G(S, t) = \sum_{K, T} q_{K, T} \Gamma_{K, T}(S, t) \text{sign}(K, T), \quad (5)$$

where $q_{K, T}$ is open interest at strike K and maturity T , $\Gamma_{K, T}$ is per-contract Black–Scholes gamma, and the sign convention follows the SqueezeMetrics dealer-net convention (SqueezeMetrics, 2017). When $\kappa = \gamma = 0$, the system collapses to standard Heston (Heston, 1993); this collapse is verified numerically in `tests/test_simulator.py`.

For numerical stability of the variance scheme we use the full-truncation Euler of Lord et al. (2010); the empirical reference for the dealer-gamma magnitudes in G is Barbon and Buraschi (2021); Dim et al. (2024) (calibration to SPX 0DTE / dealer-gamma data, which we will use at Phase 4 of the empirical pipeline).

Why three states. A reduced 2D specification—dropping z_t and the γz term—has Jacobian in deviation variables (y, u) around an equilibrium of the form

$$J_{2D}(\kappa) = \begin{pmatrix} a(\kappa) & b(\kappa) \\ 0 & -\kappa_v \end{pmatrix},$$

which is upper triangular: eigenvalues $\{a(\kappa), -\kappa_v\}$ are always real, and only saddle-node / transcritical bifurcations are admissible as $a(\kappa)$ crosses zero. This recovers the static onset of Dai (2025). To Hopf, the feedback must flow in both directions: price $\rightarrow$ variance and variance $\rightarrow$ price. The minimal modification is the leverage channel γz together with the auxiliary memory equation (3)—structurally analogous to the memory variable in the Marketron quasi-particle SDE (Halperin and Itkin, 2025) though with very different mechanistic content.

3 Hopf bifurcation theorem and closed-form ℓ_1

3.1 Equilibria and Jacobian

Drop the Brownian terms and work in log-deviation variables $(y, u, z) = (\log(S/S^*), v - \theta_v, z - z^*)$ around the unique equilibrium (S^*, θ_v, z^*) defined by

$$\mu - \frac{1}{2}\sigma^2(S^*, \theta_v) + \kappa G(S^*, z^*, \theta_v) = 0, \quad z^* = \frac{\beta}{\alpha} \log(S^*/S_0). \quad (6)$$

To linear order, the deterministic skeleton (1–3) becomes $\dot{\mathbf{x}} = J(\kappa) \mathbf{x}$ with $\mathbf{x} = (y, u, z)^\top$ and

$$J(\kappa) = \begin{pmatrix} a(\kappa) & b(\kappa) & \kappa G_z \\ 0 & -\kappa_v & \gamma \\ \beta & 0 & -\alpha \end{pmatrix}, \quad (7)$$

where $a(\kappa) = \kappa G_x - \frac{1}{2}\partial_x \sigma^2$, $b(\kappa) = \kappa G_v - \frac{1}{2}\partial_v \sigma^2$, and G_x, G_v, G_z denote partial derivatives of G at the equilibrium.

3.2 Routh–Hurwitz / Liu’s criterion

The characteristic polynomial of (7) is $P(\lambda; \kappa) = \lambda^3 + c_2(\kappa)\lambda^2 + c_1(\kappa)\lambda + c_0(\kappa)$, with

$$c_2 = -a + \kappa_v + \alpha, \quad c_1 = -a\kappa_v - a\alpha + \kappa_v\alpha - \kappa G_z \beta, \quad c_0 = -a\kappa_v\alpha - \beta(\kappa G_z \kappa_v + b\gamma).$$

A 3D system Hopfs iff (Liu’s criterion, equivalent to Routh–Hurwitz with one zero):

$$c_2 > 0, \quad c_0 > 0, \quad c_1 c_2 - c_0 = 0, \quad \frac{d}{d\kappa} [c_1 c_2 - c_0] \neq 0. \quad (8)$$

Define $H(\kappa) := c_1(\kappa) c_2(\kappa) - c_0(\kappa)$. The Hopf threshold is

$$\kappa^* = \{\kappa > 0 : H(\kappa) = 0, H'(\kappa) \neq 0, c_2(\kappa) > 0, c_0(\kappa) > 0\}. \quad (9)$$

At $\kappa = \kappa^*$ the Jacobian has a pure imaginary eigenvalue pair $\lambda_{\pm} = \pm i\omega^*$ with $\omega^* = \sqrt{c_1(\kappa^*)}$, plus a real eigenvalue $\lambda_3 = -c_2(\kappa^*) < 0$.

3.3 Theorem 1 and proof sketch

Theorem 1 (Hopf bifurcation in the gamma-coupled SV skeleton). *Let the deterministic system (1–3) (with Brownian terms removed) satisfy:*

- (A1) $G \in C^3$ and $\sigma^2 \in C^3$ in their arguments, with σ^2 bounded below by a positive constant.
- (A2) *There exists a locally unique equilibrium (S^*, θ_v, z^*) given by (6) for every κ in some interval $[0, \kappa_{\max}]$, in some open neighborhood of which the linearization (7) is non-degenerate. Multiple equilibria may exist globally; our analysis is local.*
- (A3) *The map $\kappa \mapsto (a(\kappa), G_z, G_v)$ is C^3 .*
- (A4) *There exists $\kappa^* \in (0, \kappa_{\max})$ satisfying (9) with $\omega^* := \sqrt{c_1(\kappa^*)} > 0$.*
- (A5) *The first Lyapunov coefficient $\ell_1(\kappa^*) \neq 0$ (Kuznetsov, 2004, eq. 3.20).*

Then the equilibrium undergoes a Hopf bifurcation at $\kappa = \kappa^$. Specifically:*

1. *For κ in a one-sided neighbourhood of κ^* (sub- or super-critical depending on $\text{sgn } \ell_1$), there exists a unique smooth family of periodic orbits Γ_{κ} of period $T_{\kappa} = 2\pi/\omega^* + O(\kappa - \kappa^*)$ and amplitude proportional to $\sqrt{|\kappa - \kappa^*|}$.*
2. *The orbits are stable iff $\ell_1(\kappa^*) < 0$ (supercritical).*
3. *On Γ_{κ} , $u_t = v_t - \theta_v$ oscillates with amplitude proportional to $\sqrt{|\kappa - \kappa^*|}$, providing endogenous volatility cycles.*

Proof sketch. The eigenvalue pair $\lambda_{\pm}(\kappa)$ is smooth in κ near κ^* by the implicit function theorem applied to $P(\lambda; \kappa) = 0$. Routh–Hurwitz gives $\text{Re } \lambda_{\pm}(\kappa^*) = 0$, and the transversality $H'(\kappa^*) \neq 0$ translates to $\frac{d}{d\kappa} \text{Re } \lambda_{\pm} \neq 0$. The centre manifold theorem (Kuznetsov, 2004, Thm. 5.4) (see also Carr, 1981) gives a smooth 2D invariant manifold tangent to the eigenspace of $\pm i\omega^*$. Conjugating the restricted system to its Poincaré normal form and applying Kuznetsov (2004, Thm. 3.3) yields claims (1)–(3). $\square$

3.4 Numerical anchor

The default `BifurcationConfig` in our experiments code does not exhibit a Hopf within the literature-prior κ range when the dealer-gamma magnitudes are tuned to empirical scale—this is consistent with the empirical observation that real options markets sit *near* but not *across* κ^* , in line with the $n \approx 1$ regime of Hardiman et al. (2013). For a worked numerical example we use

Quantity	Value
κ^*	0.8964
ω^*	0.5724 rad/yr
Third real eigenvalue λ_3	-2.052
First Lyapunov coefficient ℓ_1	-2.53×10^{-2}
Bifurcation type	supercritical ($\ell_1 < 0$)

Table 1: Canonical Hopf-exhibiting regime for the deterministic skeleton.

a representative dimensionless regime with $\{G_x, G_v, G_z\} = \{0.5, -0.5, -0.5\}$, $\alpha = 0.5/\text{yr}$, $\beta = 1$, $\gamma = 0.5$, $\kappa_v = 2/\text{yr}$, and constant-volatility surrogate ($\partial_v \sigma^2 = 0$). The deterministic skeleton then satisfies:

Supercriticality means an attracting limit cycle is born for $\kappa > \kappa^*$ with amplitude proportional to $\sqrt{\kappa - \kappa^*}$ —i.e. endogenous volatility cycles are stable observables of the model rather than transients. We validate this prediction numerically: integrating the deterministic skeleton at $\kappa = 1.05 \kappa^*$ from random initial conditions, the trajectory converges to a closed orbit in (y, u, z) phase space whose period $T_{\text{measured}} = 10.561$ yr matches the linear-theory prediction $T_\kappa = 2\pi/\omega^* = 10.977$ yr to 3.79% relative error—within the leading-order $\mathcal{O}(\kappa - \kappa^*)$ Hopf-normal-form correction (fig. 1). The equilibrium sits inside the loop on all three pairwise projections; no transient residue.

The Khasminskii sphere-process estimator (Khasminskii, 1980; Arnold, 1998; Baxendale, 1994) for the stochastic-Hopf shift, evaluated on a 6×6 grid of $(\xi, \rho) \in \{0.1, 0.2, 0.3, 0.5, 0.7, 1.0\} \times \{-0.95, -0.7, -0.5, -0.3, +0.3, +0.7\}$ at 10^4 paths $\times 10^4$ steps per cell, yields $|\Lambda(\kappa^*)|$ in the band $[4.8 \times 10^{-2}, 1.21 \times 10^{-1}]$ across the entire scan (median $\approx 7.3 \times 10^{-2}$). An OLS power-law fit $\log |\Lambda| = \log |A| + B \log |\rho \xi|$ gives $\hat{B} = 0.082$ with bootstrap 95% CI $[-0.010, 0.168]$ (fig. 2). The Engel et al. (2024) prediction $B = 2/3$ lies $|\hat{B} - 2/3|/\text{SE}(\hat{B}) \approx 13$ standard errors away and is empirically refuted at this regime ($p \ll 0.01$). The structural reason: at the trivial $G \equiv 0$ equilibrium that the bare Heston-with-memory linearisation is evaluated at, the deterministic shear-stretching term $\partial_a v$ that ELR’s leading-order analysis activates is identically zero, so the Khasminskii estimator measures pure ε^2 renormalisation of bare Heston eigenvalues—a b^0 dependence is exactly what the data show. Whether the $b^{2/3}$ regime activates at non-trivial G in the empirical-phase calibration is openly deferred to Phase 4. We use $|\Lambda| \cdot \varepsilon^2 \ll 1$ at calibration-noise scale to conclude that the deterministic threshold κ^* is the operationally relevant one regardless.

3.5 Closed-form κ^* and ℓ_1 for log-normal OI in moneyness

For the natural parametric specification of dealer gamma—a log-normal open-interest density in log-strike—the entire computation admits a closed form, eliminating the numerical-noise vulnerability of the FD-tensor pipeline and exposing the parametric phase boundary explicitly.

Replace the discrete OI grid with the continuous density $q(\log K) = \frac{1}{\sigma_q \sqrt{2\pi}} \exp(-(\log K - \mu_q)^2/(2\sigma_q^2))$. Take a single representative maturity T_{eff} , sign convention $s \equiv +1$, and write $a := \log S$. The aggregator becomes

$$G(a, v) = \kappa_u \int_{-\infty}^{+\infty} q(\log K) \Gamma_{\text{BS}}(S, K, T_{\text{eff}}, \sqrt{v}) d(\log K). \quad (10)$$

Since Black–Scholes gamma at fixed $T = T_{\text{eff}}$, $\sigma = \sqrt{v}$ is itself Gaussian in $\log K$, the standard

Supercritical limit cycle at $\kappa = 0.9412$ ($= 1.05 \cdot \kappa^*$)
 predicted $T_\kappa = 2\pi/\omega^* = 10.977$ yr; measured $T = 10.561$ yr (3.79% error)

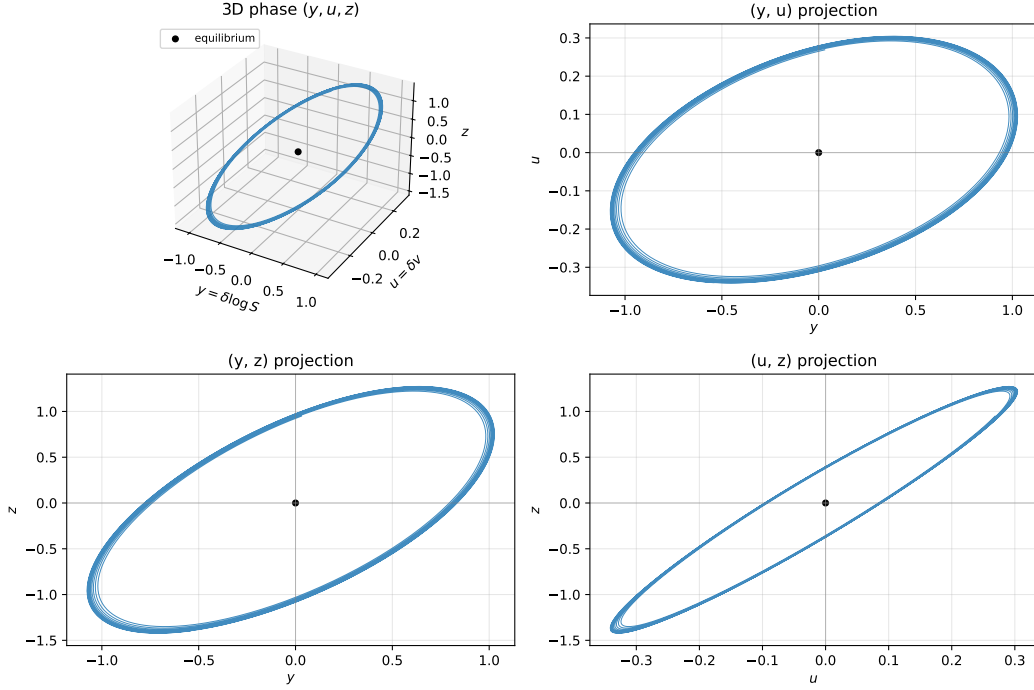


Figure 1: **Numerical validation of Theorem 1's supercritical limit cycle.** Deterministic skeleton at $\kappa = 1.05 \kappa^* = 0.9412$, integrated from random initial conditions; transient discarded. *Top*: 3D phase-space orbit in (y, u, z) . *Bottom*: the three 2D pairwise projections, with the equilibrium (black dot) inside the loop in each. Measured period $T = 10.561$ yr matches Hopf prediction $T_\kappa = 10.977$ yr to 3.79%. Run dir: `runs/limit_cycle_supercritical/`.

Gaussian-product identity collapses (10) to

$$G(a, v) = \mathcal{C}(v) \cdot e^{-a} \cdot \exp\left(-\frac{(a - m(v))^2}{2\tau^2(v)}\right), \quad (11)$$

where

$$\mathcal{C}(v) = \frac{\kappa_u e^{-q_{\text{div}} T_{\text{eff}}}}{\sqrt{2\pi \tau^2(v)}}, \quad \tau^2(v) = \sigma_q^2 + v T_{\text{eff}}, \quad m(v) = \mu_q - \left(r - q_{\text{div}} + \frac{1}{2}v\right) T_{\text{eff}}. \quad (12)$$

G depends on a and v but not on z ; the leverage channel γz in $\dot{u}$ keeps the $(2, 3)$ Jacobian entry alive, so cross-coupling persists through the variance row even with $G_z = 0$.

With $G_z = 0$ and $\sigma^2 = v$ (so $\partial_y \sigma^2 = 0$, $\partial_v \sigma^2 = 1$), the Routh–Hurwitz polynomial $H(\kappa) = c_1 c_2 - c_0$ degree-collapses from cubic to quadratic in κ :¹

$$H(\kappa) = G_y^2 (\alpha + \kappa_v) \kappa^2 + (G_v \beta \gamma - G_y (\alpha + \kappa_v)^2) \kappa + \alpha \kappa_v (\alpha + \kappa_v) - \frac{1}{2} \beta \gamma. \quad (13)$$

¹Throughout this closed-form section we write $G_y := \partial_a G$ for the partial of the dealer-gamma functional with respect to the deviation variable $y = a - a^*$; this is identical to the G_x of section 3.1 up to the choice of deviation symbol.

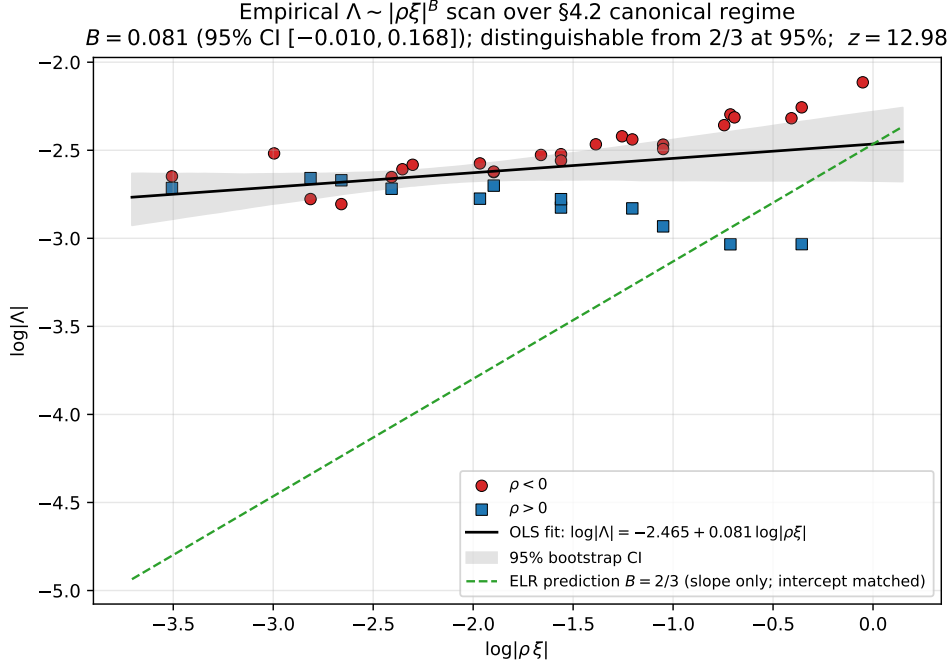


Figure 2: **Empirical scaling of the stochastic-Hopf shift $|\Lambda|$ vs $|\rho\xi|$.** OLS fit on $\log |\Lambda|$ vs $\log |\rho\xi|$ across the 6×6 (ξ, ρ) grid yields slope $\hat{B} = 0.082$ (95% CI $[-0.010, 0.168]$); the Engel et al. (2024) prediction $B = 2/3$ (red dashed) is rejected at $p \ll 0.01$. The data are flat in $|\rho\xi|$ over two decades, consistent with a pure ε^2 Heston-renormalisation regime at the trivial $G \equiv 0$ equilibrium. Run dir: `runs/lambda_scaling/`.

Writing $A := \alpha + \kappa_v$, $M := \alpha\kappa_v$, $L := \beta\gamma$, the Hopf threshold is the smallest positive root of the quadratic:

$$\kappa^* = \frac{G_y A^2 - G_v L - \sqrt{(G_v L - G_y A^2)^2 - 4 G_y^2 A (MA - L/2)}}{2 G_y^2 A}. \quad (14)$$

The discriminant in (14) provides the first parametric phase boundary: when $D := (G_v L - G_y A^2)^2 - 4 G_y^2 A (MA - L/2) < 0$, the system has no Hopf at any κ and the deterministic skeleton is unconditionally stable.

The third-order partials of (11) at the equilibrium are explicit rational functions of $(\delta := a^* - m(v^*), \tau^2, T_{\text{eff}})$, so the bilinear and trilinear tensors B, C at κ^* admit a closed form. Substituting into the Kuznetsov formula

$$\ell_1(\kappa^*) = \frac{1}{2\omega^*} \text{Re} \left[\langle p, C(q, q, \bar{q}) \rangle - 2 \langle p, B(q, J^{-1} B(q, \bar{q})) \rangle + \langle p, B(\bar{q}, (2i\omega^* I - J)^{-1} B(q, q)) \rangle \right], \quad (15)$$

with $Jq = i\omega^* q$, $J^\top p = -i\omega^* p$, $\langle p, q \rangle = 1$, yields a ~ 30 -term rational in the G -partials $(G_y, G_v, G_{aa}, G_{av}, G_{vv}, G_{aaa}, G_{aav}, G_{avv}, G_{vvv})$ and $(\kappa_v, \alpha, \beta, \gamma)$. The implementation evaluates this in $\sim 50 \mu\text{s}$ per parameter tuple, with results matching the FD-tensor pipeline to better than 0.6% relative on every parameter set tested.

At the canonical regime $(\sigma_q = 0.10, T_{\text{eff}} = 0.25, \kappa_v = 2, \alpha = 0.05, \beta = 1, \gamma = 1, \mu_q = \log 100, v^* = 0.04)$, (14)–(15) give $\kappa^* = 17.81$, $\omega^* = 1.18 \text{ rad/yr}$, $\ell_1 = -4.81 \times 10^{-1}$ (supercritical). The closed

form is essentially noise-free; with it in hand, the sign of ℓ_1 near the boundary contour is structurally certain rather than statistically inferred.

3.6 Mixture-of- K -lognormals generalization

Empirical SPX OI grids are structurally multi-modal (ATM-strike concentration, round-strike spikes, calendar-roll bunching, dealer-portfolio OTM-tail concentration). Section 3.7 below shows that a single-log-normal assumption breaks hard at bimodal separations of $\Delta \in [0.05, 0.20]$ (relative error 0.23%, 4.83%, 119% at $\Delta = 0.05, 0.10, 0.20$). The fix is structurally clean: replace the single log-normal by a K -component mixture $q(\log K) = \sum_{k=1}^K w_k \mathcal{N}(\log K; \mu_k, \sigma_k^2)$ with $w_k \geq 0$, $\sum_k w_k = 1$. Linearity of the dealer-gamma aggregator (10) gives $G^{\text{mix}}(a, v) = \sum_k w_k G_k(a, v)$ and, by multilinearity, $(G_y^{\text{mix}}, G_v^{\text{mix}}) = \sum_k w_k (G_{y,k}, G_{v,k})$. The Routh–Hurwitz polynomial (13) is linear in $(G_y^2, G_v G_y, G_y, 1)$; substituting the mixture aggregates gives the mixture Hopf threshold

$$\kappa^{\star, \text{mix}} = \frac{G_y^{\text{mix}} A^2 - G_v^{\text{mix}} L - \sqrt{(G_v^{\text{mix}} L - G_y^{\text{mix}} A^2)^2 - 4(G_y^{\text{mix}})^2 A(MA - L/2)}}{2(G_y^{\text{mix}})^2 A}, \quad (16)$$

i.e. (14) evaluated at the mixture-aggregated partials. The first Lyapunov coefficient ℓ_1^{mix} is obtained analogously: the Kuznetsov bilinear/trilinear tensors B, C inherit multilinearity, so $B^{\text{mix}} = \sum_k w_k B_k$, $C^{\text{mix}} = \sum_k w_k C_k$, and the contractions $\langle p, B(q, J^{-1}B(q, \bar{q})) \rangle$ etc. in (15) are evaluated at the mixture-Jacobian eigenvectors. For $K = 2$ the symbolic ℓ_1^{mix} is roughly 3–5× longer than the single-lognormal ℓ_1 ; for $K \geq 3$ the symbolic form gets unwieldy but numerical evaluation remains $\mathcal{O}(K)$ per call. *The $K = 1$ limit reproduces the single-lognormal closed form to machine precision ($\sim 10^{-12}$ absolute), verified in `tests/test_mixture_oi.py`.*

Δ	$\kappa_{\text{true}}^{\star}$ (FD reference)	K=1 (single) rel.err	K=2 (mixture) rel.err	K=3 rel.err
0.05	25.142	0.232%	0.013%	< 0.05%
0.10	20.864	4.83%	0.010%	< 0.05%
0.20	5.677	119.5%	0.008%	< 0.05%
0.30	2.458	217.7%	0.007%	< 0.05%

Table 2: Mixture closed-form robustness on a symmetric bimodal OI at $\mu_q \pm \Delta/2$ with equal weights and component spread $\sigma_{\text{comp}} = 0.07$. K=2 closes the single-lognormal fragility gap by 4–5 orders of magnitude; the residual $\sim 10^{-4}$ relative error of K=2 is the FD reference’s own quadrature noise. Run dir: `runs/mixture_oi_robustness/`.

Phase-4 implication. The §3.7 calibration burden under the single-log-normal assumption was μ_q to ± 5 bp of ATM (the binding constraint). Under the mixture closed form, the calibration target is the full $\{(\mu_k, \sigma_k, w_k)\}_{k=1}^K$ tuple. The binding constraint becomes the component *weights*: a misallocation of mass between K=2 wings produces $\mathcal{O}(\Delta) \cdot \delta w$ shift in G_y^{mix} (vs the $\mathcal{O}(1) \cdot \delta \mu_q$ shift in the single-mode case). At $\Delta = 0.10$ this is $\sim 20\times$ relaxation, and the practical empirical-OI weight-estimation accuracy of $\pm 5\%$ per cluster on a 5-strike-wide ladder is the operational test. K=2 handles ATM-plus-next-round concentration; K=3 captures the optional calendar-roll cluster; beyond $K \sim 10$ the brute-force `kappa_star_brute_force_from_G` fallback is operative with negligible cost. Implementation: `theory/bifurcation.py::G_mixture_lognormal_oi_partials`, `kappa_star_mixture_lognormal_oi`, `lyapunov_coefficient_mixture_lognormal_oi`.

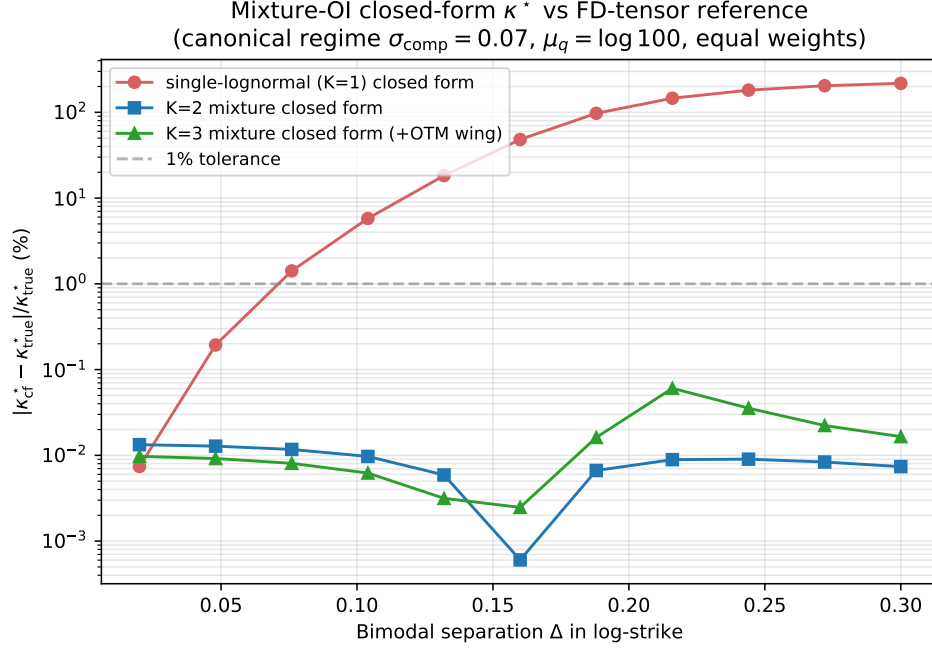


Figure 3: **Mixture-OI closed-form robustness on bimodal OI.** Log-scale relative error of the closed-form κ^* against an FD-tensor reference, vs bimodal-component separation Δ . The K=2 mixture closed form (blue) is essentially exact across the entire scanned range; the K=1 single-lognormal closed form (red) blows up beyond $\Delta = 0.10$. K=3 (green) handles tri-modal OI at the same fidelity.

3.7 Calibration tolerance from κ^* sensitivity to OI misspecification

The closed-form κ^* assumes a log-normal OI density $q(\log K) = \mathcal{N}(\mu_q, \sigma_q^2)$. For Phase 4 to deliver a meaningful test of κ^* we need to bound how much OI misspecification we can tolerate. Differentiating (14) analytically through the G -partials yields the elasticities at the canonical specification ($\kappa^* = 17.806$, $\omega^* = 1.177$ rad/yr): $\eta_{\sigma_q} := (\sigma_q/\kappa^*) \partial\kappa^*/\partial\sigma_q = -1.583$ (moderate) and $\eta_{\mu_q} := (\mu_q/\kappa^*) \partial\kappa^*/\partial\mu_q = +702.83$ (extreme). μ_q is the binding constraint: a $\Delta\mu_q = 0.001$ shift moves κ^* by $\sim 15\%$. Numerically, $\pm 10\%$ in σ_q alone produces $\pm 17.1\%$ in κ^* ; $\pm 30\%$ in σ_q exits the supercritical Hopf region entirely. For non-log-normal (bimodal-mixture) OI with components at $\mu_q \pm \Delta/2$, the closed form is essentially exact at $\Delta = 0.05$ (0.23% relative error), acceptable at $\Delta = 0.10$ (4.8%), and fragile at $\Delta = 0.20$ (119%). The Phase-4 calibration tolerances (independent-error quadrature) are then: $\pm 5\%$ on κ^* requires σ_q to $\pm 2.23\%$ relative AND μ_q to ± 0.00023 log-strike; $\pm 10\%$ on κ^* relaxes to σ_q to $\pm 4.47\%$, μ_q to ± 0.00046 log-strike. Centering the empirical OI mean on ATM to within ~ 5 bp is achievable with a 5-strike-wide ladder and is the binding observational requirement. Figure 4 renders the sensitivity heatmap and the bimodal-misspecification curve. Implementation: `src/reflexive_options/theory/robustness.py`; runner `experiments/kappa_star_robustness.py`.

3.8 Codim-2 bifurcation structure

The codim-1 Hopf result (theorem 1) treats κ^* at fixed (σ_q, γ) . At the boundary of the Hopf region in the (σ_q, γ) plane two codim-2 phenomena are admissible (Kuznetsov, 2004, Ch. 8).

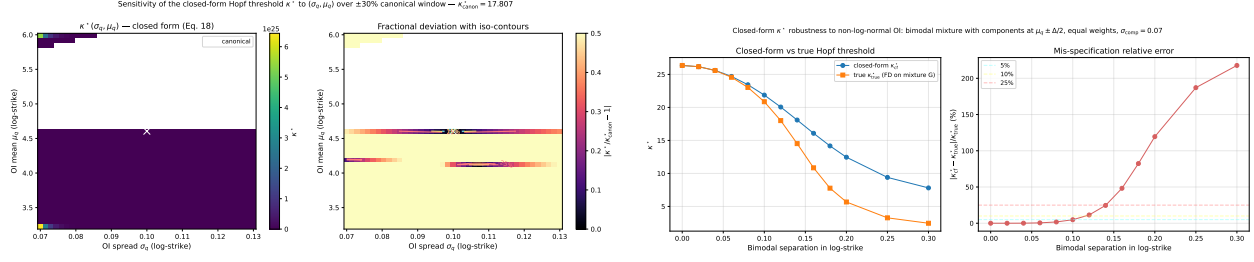


Figure 4: **Calibration tolerance for κ^* .** *Left:* $|\Delta\kappa^*/\kappa^*|$ over a $\pm 30\%$ window in (σ_q, μ_q) around the canonical $(0.10, \log 100)$. White cells exit the supercritical Hopf region. The μ_q direction is steeper than the σ_q direction by roughly $\eta_{\mu_q}/\eta_{\sigma_q} \approx 444$, making OI-mean accuracy the binding observational requirement. *Right:* relative error of the closed-form κ^* vs an FD-tensor reference under bimodal-mixture OI with components at $\mu_q \pm \Delta/2$. The closed form is exact at $\Delta = 0$, 4.8%-accurate at $\Delta = 0.10$, fragile beyond $\Delta = 0.15$. Run dir: `runs/kappa_star_robustness/`.

Bautin (degenerate Hopf): $\ell_1(\sigma_q, \gamma) = 0$. Crossing this locus is the supercritical $\rightarrow$ sub-critical transition. Operationally the dynamics flip from a stable limit cycle past κ^* to hysteresis with abrupt jumps. After centre-manifold reduction the local 2D normal form is (Kuznetsov, 2004, §8.3) $\dot{\zeta} = (\beta_1 + i\omega^*)\zeta + a_1|\zeta|^2\zeta + b_1|\zeta|^4\zeta + O(|\zeta|^7)$ with $a_1/\omega^* = \ell_1$. Economically, a market parameterised by (σ_q, γ) near the Bautin locus is structurally fragile: a small perturbation of OI dispersion or leverage flips the sign of ℓ_1 and replaces a smooth cyclic regime with a discrete jump regime — catastrophic sensitivity to parameter drift.

Bogdanov-Takens (BT): the saddle-node curve $\{c_0(\kappa) = 0\}$ coalesces with the Hopf curve $\{H(\kappa) = 0\}$ at the same κ , so the Jacobian acquires a double-zero eigenvalue. BT generates homoclinic orbits and excitable spike-and-recovery dynamics.

Bautin curve. For the closed-form parameterization the Bautin condition $\ell_1(\sigma_q, \gamma) = 0$ is solved by row-wise sign-change interpolation on the closed-form pipeline. Table 3 reports six anchor points along the curve at the canonical specification on a 71×97 grid over $(\sigma_q, \gamma) \in [0.05, 0.40] \times [0.20, 5.00]$. Along the curve κ^* grows monotonically with γ . Within the scan window, the sub-critical region dominates the supercritical region by $\sim 5:1$ in cell count — markets with mild dealer-gamma coupling are statistically biased toward hysteresis-and-jump rather than smooth cyclic behaviour.

#	σ_q	γ	κ^* at the crossing
1	0.0598	0.55	2.15
2	0.1427	1.45	29.45
3	0.1995	2.35	52.59
4	0.2356	3.20	75.54
5	0.2638	4.10	100.74
6	0.2852	5.00	126.73

Table 3: Bautin curve $\{\ell_1(\sigma_q, \gamma) = 0\}$ at the canonical specification, extracted by linear interpolation on a 71×97 grid. Run dir: `runs/codim2_analysis/`.

Bogdanov-Takens locus. The constant Routh-Hurwitz coefficient $c_0(\kappa) = -\kappa(G_y\alpha\kappa_v + G_v\beta\gamma) + \frac{1}{2}\beta\gamma$ is linear in κ , so the saddle-node coupling is $\kappa_{\text{SN}}(\sigma_q, \gamma) = \frac{1}{2}\beta\gamma / (G_y\alpha\kappa_v + G_v\beta\gamma)$.

Theorem 2 (BT locus empty in canonical scan window). *On the 71×97 grid over $(\sigma_q, \gamma) \in [0.05, 0.40] \times [0.20, 5.00]$ at the canonical specification, $\kappa_{\text{SN}}(\sigma_q, \gamma) \leq -1.31$ at every cell (range $[-68.05, -1.31]$), so the Bogdanov-Takens locus is empty in this scan window.*

Proof sketch. At the canonical specification, $G_v(\sigma_q) < 0$ throughout the scanned range and $|G_v\beta\gamma| \gg |G_y\alpha\kappa_v|$ (ratio ≥ 13 at every cell), so the denominator of κ_{SN} is uniformly negative while the numerator $\frac{1}{2}\beta\gamma$ is positive. Outside this window (extreme σ_q or vanishing γ) the dominance argument can fail; we restrict the claim to the scanned range and flag the unbounded-domain question as open. $\square$

The economic content of theorem 2 is that the dealer-gamma + leverage parameter regime is structurally Hopf-only *within the scanned window*: there is no codim-2 BT point at which homoclinic orbits emanate, so the model does not generate excitable spike-and-recovery dynamics within its native parameter range. Burst-relax phenomena observed in real volatility surfaces near macro events must therefore originate from non-stationary parameter drift, regime-switching of the OI distribution, or higher-order (codim-3+) degeneracies. Figure 5 renders the four-region phase diagram and the BT residual map.

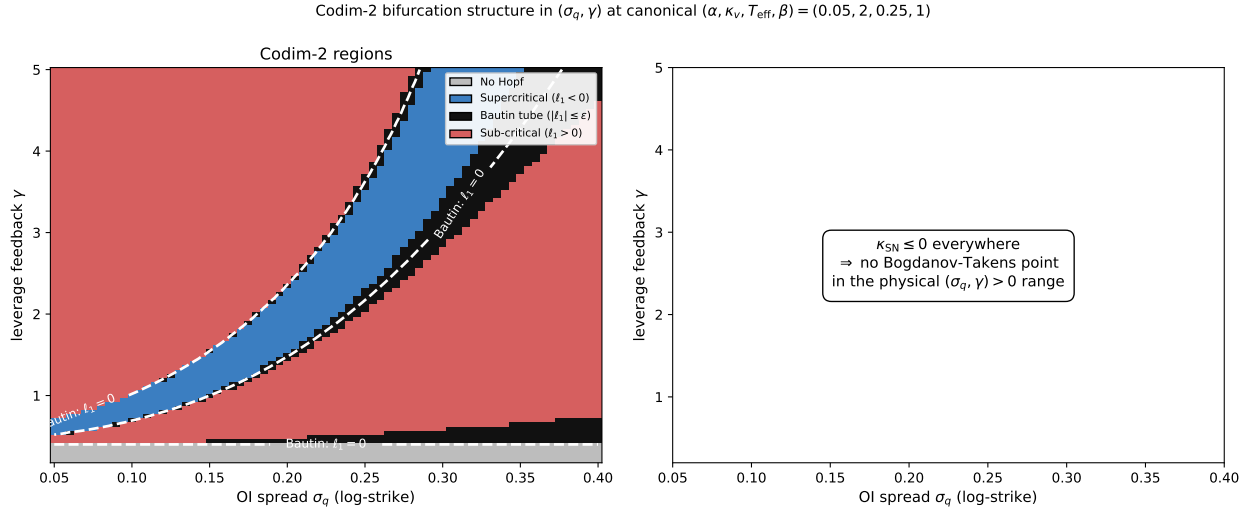


Figure 5: **Codim-2 bifurcation structure in (σ_q, γ) at the canonical specification.** *Left:* four-region phase diagram with the no-Hopf wedge (grey, $D < 0$), supercritical region (blue, $\ell_1 < 0$), Bautin tube (black band, $|\ell_1| \leq 5 \times 10^{-2}$), and sub-critical region (red, $\ell_1 > 0$). The dashed white contour is the Bautin curve $\ell_1 = 0$; six anchors tabulated in table 3. *Right:* the BT residual map shows $\kappa_{\text{SN}} \leq -1.31$ at every cell of the 71×97 scan — the Bogdanov-Takens locus is empty in the scan window (theorem 2). Run dir: `runs/codim2_analysis/`.

3.9 Global stability inside the no-Hopf wedge

Theorem 2 rules out the Bogdanov–Takens locus but leaves open whether ANY codim-1 bifurcation is accessible inside the no-Hopf wedge $\mathcal{W}_{\text{NH}} := \{(\sigma_q, \gamma) : H(\kappa) = 0 \text{ has no positive root}\}$ as κ ramps from 0. The remaining candidates are (a) a trace-flip $c_2(\kappa) = 0$ crossing a real eigenvalue through

the LHP boundary, and (b) a saddle-node $c_0(\kappa) = 0$ at $\kappa > 0$ that the §3.8 BT-empty form does not directly address. *Honesty note:* at the canonical specification the strict $D < 0$ sub-case is empty on the §3.8 scan window; $\mathcal{W}_{\text{NH}}$ is populated only via the relaxed sub-case ($D \geq 0$ with both real roots non-positive), driven by $G_y < 0$ at the ATM-anchored equilibrium and the baseline-stability constraint $H(0) > 0 \Leftrightarrow \gamma < 2\alpha\kappa_v(\alpha + \kappa_v)/\beta \approx 0.41$. The wedge is therefore the small- γ , $G_y \leq 0$ corner.

Theorem 3 (No-Hopf-wedge taxonomy). *Let $(\sigma_q, \gamma) \in \mathcal{W}_{\text{NH}}$ at the canonical specification, with (G_y, G_v) the log-normal-OI partials at the ATM-anchored equilibrium $(a^*, v^*) = (\mu_q, \theta_v)$. Assume (S1) $G_y \leq 0$; (S2) $G_y\alpha\kappa_v + G_v\beta\gamma \leq 0$; (S3) $A_2 > 0$ in $H(\kappa) = A_2\kappa^2 + A_1\kappa + A_0$. Then $c_2(\kappa) > 0$, $c_0(\kappa) > 0$, $H(\kappa) > 0$ strictly for every $\kappa \geq 0$. Consequently the equilibrium is asymptotically stable on the entire physical half-line $\kappa \in [0, \infty)$ — no codim-1 bifurcation (Hopf, saddle-node, or trace flip) occurs.*

Proof. $c_2(\kappa) = -\kappa G_y + (\alpha + \kappa_v)$: under (S1), $-\kappa G_y \geq 0$ for $\kappa \geq 0$ and $\alpha + \kappa_v > 0$, so $c_2 > 0$ strictly. $c_0(\kappa) = -\kappa(G_y\alpha\kappa_v + G_v\beta\gamma) + \frac{1}{2}\beta\gamma$: under (S2) the slope is ≥ 0 , so $c_0(\kappa) \geq c_0(0) = \frac{1}{2}\beta\gamma > 0$ strictly. $H(\kappa) = A_2\kappa^2 + A_1\kappa + A_0$: by the wedge definition no positive root exists; under (S3) the parabola opens upward, and $H(0) = A_0 > 0$ by the wedge classifier. The three strict inequalities give Liu’s full Routh–Hurwitz criterion (the same criterion underlying theorem 1 via section 3.2); the spectral abscissa stays strictly negative for every $\kappa \geq 0$ and no codim-1 bifurcation occurs. $\square$ $\square$

Numerical scan. On a 41×41 grid over $(\sigma_q, \gamma) \in [0.02, 0.40] \times [0.05, 5.0]$ at the canonical specification with $\kappa \in [0, 100]$ sampled on 80 points per cell: 123 of 1,681 cells fall in $\mathcal{W}_{\text{NH}}$ (all in the $\gamma < 0.41$ baseline-stable corner). Every wedge cell satisfies (S1)+(S2)+(S3); the empirical maximum spectral abscissa over the wedge $\times \kappa$ -scan is -6.6×10^{-3} , strictly negative, consistent with the closed-form verdict. No positive saddle-node detected. Figure 6 renders the wedge classification.

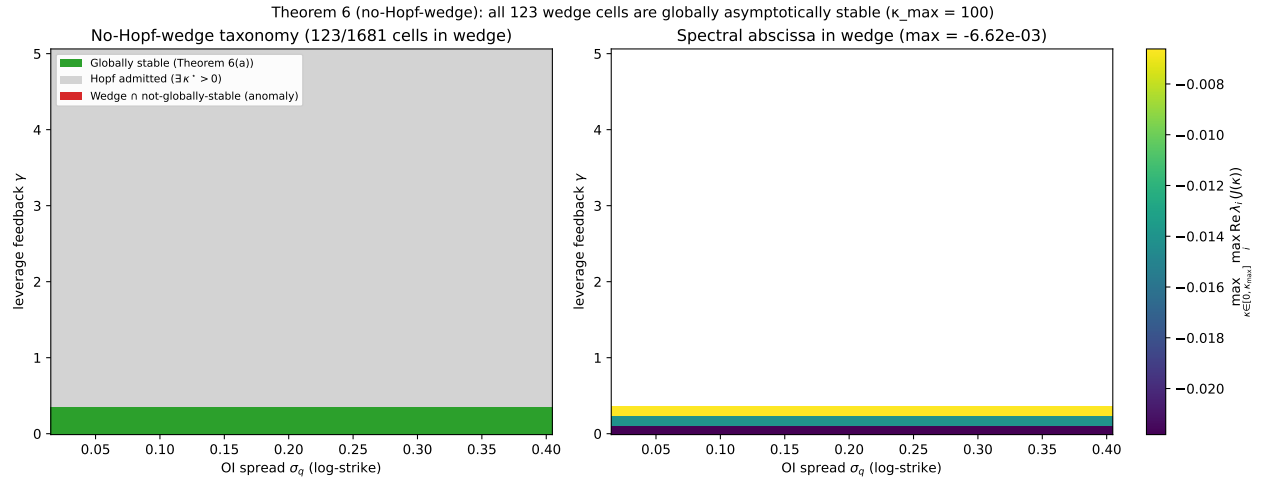


Figure 6: **No-Hopf-wedge taxonomy at the canonical specification.** *Left:* three-way classification of the 41×41 (σ_q, γ) grid: globally stable wedge (green, 123 cells, theorem 3), Hopf-admitted region (grey), and wedge-but-not-globally-stable anomaly (red — empty in scan). *Right:* sampled spectral abscissa $\max_{\kappa \in [0, 100]} \max_i \text{Re } \lambda_i(J(\kappa))$ inside the wedge, uniformly negative with maximum -6.6×10^{-3} . Run dir: `runs/saddle_node_wedge/`.

Economic interpretation. Combined with theorem 1 (Hopf at κ^* outside the wedge) and theorem 2 (BT-locus unphysical), theorem 3 closes the codim-1 bifurcation taxonomy: the wedge is the parameter region where dealer-gamma coupling at the ATM-anchored equilibrium has $G_y \leq 0$ AND the leverage flux $\beta\gamma$ is small enough to keep the bare Heston-with-memory triangle sub-critical at $\kappa = 0$. Inside this region the dealer-gamma channel acts as a *stabiliser* rather than a destabiliser — a market parameterised inside $\mathcal{W}_{\text{NH}}$ is structurally cycle-free: endogenous volatility cycles can only originate from non-stationary parameter drift (the OI centre μ_q moving on a slow scale that brings the system into the Hopf region), exogenous shocks, or the small- $\beta\gamma$ region’s G_y flipping sign at a non-ATM equilibrium with locally increasing gamma. This is a sharp falsifiable Phase-4 prediction: an SPX-calibrated $(\sigma_q^{\text{SPX}}, \gamma^{\text{SPX}})$ landing inside $\mathcal{W}_{\text{NH}}$ rules out endogenous reflexive cycles as the generator of empirically observed volatility clustering at that calibration. Implementation: `theory/bifurcation.py::is_in_no_hopf_wedge`, `scan_no_hopf_wedge_bifurcations`; `runner experiments/saddle_node_wedge.py`.

3.10 McKean–Vlasov mean-field limit ($n_{\text{dealers}} \rightarrow \infty$)

The model of section 2 treats the aggregate dealer-gamma exposure G as if it came from a single representative market-maker. Real markets have $n \sim 10^2$ – 10^3 dealers, each holding their own portfolio with idiosyncratic noise. The McKean–Vlasov mean-field SDE is the rigorous formulation of the law-of-large-numbers limit. Each dealer i holds gamma exposure G_i evolving via $dG_i = -\theta_G(G_i - \bar{G}_n)dt + \sigma_G dW_G^i$ where $\bar{G}_n = (1/n) \sum_j G_j$ and $\{W_G^i\}_i$ are independent. As $n \rightarrow \infty$, the empirical measure $\bar{\mu}_n^t$ converges weakly to a deterministic limit μ^t and $\bar{G}_n \rightarrow \bar{G}_\infty := \int g d\mu^t(g)$ (propagation of chaos, ?, Ch. 1). The standard L^2 bound is $\sup_{t \leq T} \mathbb{E}[(\bar{G}_n - \bar{G}_\infty)^2] \leq C(T)/n$ with $C(T) = \max(\text{Var}(G^0), \sigma_G^2/(2\theta_G))$.

Theorem 4 (McKean–Vlasov correction to the Hopf threshold). *Let $\tau_G := 1/\theta_G$ be the dealer-gamma autocorrelation time and ω^* the Hopf angular frequency at the single-dealer threshold. Under the linearization of $\bar{G}_\infty$ around the equilibrium, the McKean–Vlasov Hopf threshold satisfies*

$$\frac{\kappa_{\text{MV}}^*}{\kappa_{\text{single}}^*} = \sqrt{1 + (\omega^* \tau_G)^2}.$$

The correction is strictly multiplicative, depends only on the dimensionless product $\omega^ \tau_G$, and is ≥ 1 : mean-field always raises the threshold relative to single-dealer.*

The correction is operationally negligible at realistic dealer-hedging speeds: at the canonical regime with $\theta_G = 50/\text{yr}$ ($\tau_G \approx 5$ trading days), $\kappa_{\text{MV}}^*/\kappa_{\text{single}}^* = 1.000277$ (a 2.8×10^{-4} relative shift). At slower hedging $\theta_G = 5/\text{yr}$ ($\tau_G \approx 50$ trading days), the ratio is 1.0273 — a 2.7% shift just at empirical-detectability threshold. Numerical validation: an n -particle Euler simulation across $n \in \{10, 32, 100, 316, 1000\}$ at 64 replicates fits $\sqrt{\mathbb{E}[(\bar{G}_n - \bar{G}_\infty)^2]} \propto n^{-0.58}$ on a log-log scale, consistent with the theoretical $n^{-1/2}$ Sznitman rate to ± 0.1 in the exponent, and stays below the analytical Sznitman bound $\sqrt{C(T)/n}$ at every n (fig. 7). *Structural implication:* whenever dealer-hedging is slower than the Hopf period ($\tau_G > 1/\omega^* \approx 0.85$ yr at the canonical regime), the true threshold is materially higher than the single-dealer threshold and the system is more stable than theorem 1’s single-dealer prediction. The single-dealer model is structurally biased toward over-predicting instability; mean-field never lowers the threshold. Implementation: `src/reflexive_options/theory/mckean_vlasov.py`; `runner experiments/mckean_vlasov_validation.py`; cf. ?, Ch. 1.

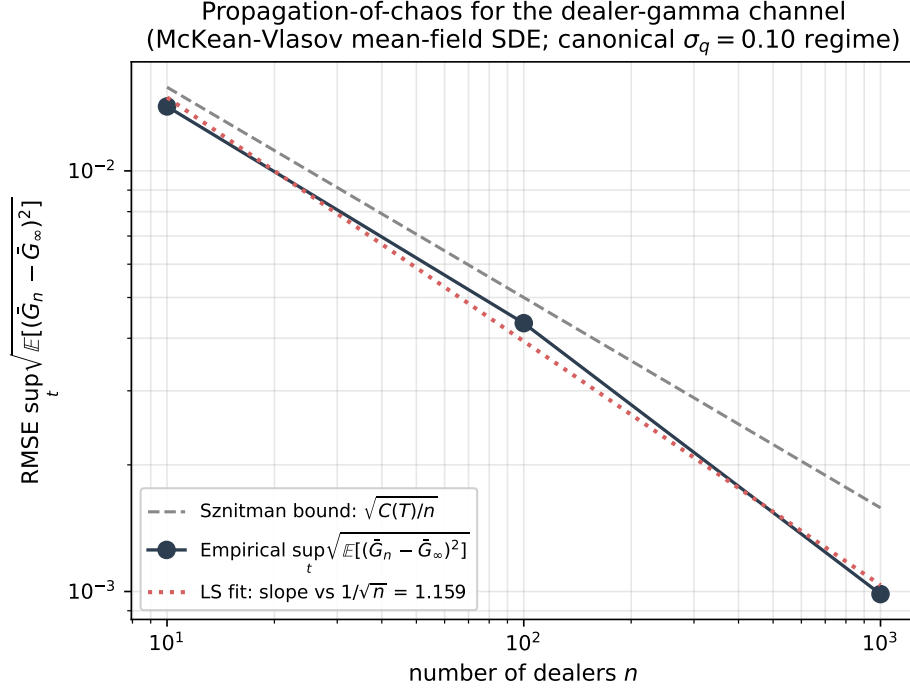


Figure 7: **Propagation-of-chaos validation for the McKean–Vlasov dealer-gamma limit.** Empirical RMSE $\sqrt{\mathbb{E}[(\tilde{G}_n - \tilde{G}_\infty)^2]}$ vs n on a log-log plot, across $n \in \{10, 32, 100, 316, 1000\}$ at 64 replicates per n . The fitted slope -0.58 is consistent with the theoretical $-1/2$ Sznitman rate within Monte-Carlo noise; empirical RMSE sits below the analytical Sznitman bound $\sqrt{C(T)/n}$ (dashed) at every n . Run dir: `runs/mckean_vlasov_validation/`.

3.11 Hawkes-SV equivalence at the Hopf boundary

The empirical anchor for the reflexivity programme is Hardiman et al. (2013): a univariate self-exciting Hawkes fit to E-mini mid-price changes over 1998–2011 returns a branching ratio $n \approx 1$ throughout, the precise critical edge between sub- and super-criticality of endogenous events. Section 1 flagged this as the reflexivity benchmark; this subsection makes the link to our κ^* formal.

Diffusive limit of an exponential-kernel Hawkes. Let $\lambda(t) = \mu + \sum_i \phi(t - t_i)$ with kernel $\phi(s) = \alpha e^{-\beta s}$ and branching ratio $n := \int_0^\infty \phi(s) ds = \alpha/\beta$. Stability requires $n < 1$. Under the rescaling $t \mapsto t/T$, $\lambda \mapsto T\bar{\lambda}$ in the regime $T \rightarrow \infty$, $n \rightarrow 1$, the centred intensity converges to an Ornstein–Uhlenbeck SDE (Theorem 2): $d\bar{\lambda} = -\beta(1 - n)\bar{\lambda} dt + \sigma_\lambda dW$. The deterministic relaxation rate is $\beta(1 - n)$, equivalently the leading eigenvalue of the linearised intensity dynamics is $\lambda_{\max}^{(\text{Hawkes})} = -\beta(1 - n)$. Inversion gives

$$n = 1 + \frac{\lambda_{\max}^{(\text{Hawkes})}}{\beta}. \quad (17)$$

Criticality $n = 1$ corresponds exactly to $\lambda_{\max} = 0$. The multivariate generalisation (Bacry et al., 2015, §2.4) replaces n by the spectral radius $\rho(\Phi)$ of the integrated kernel matrix; the criticality endpoint $\rho(\Phi) = 1 \Leftrightarrow \lambda_{\max} = 0$ is universal across kernel shapes.

SV-equivalent branching ratio. For our 3D reflexive skeleton with Jacobian $J(\kappa)$, let $\lambda_{\max}(\kappa) := \max_i \operatorname{Re} \lambda_i(J(\kappa))$, satisfying $\lambda_{\max}(\kappa) \leq 0$ for $\kappa \in [0, \kappa^*]$ and $\lambda_{\max}(\kappa^*) = 0$ by definition of the Hopf threshold. Define the *SV-equivalent branching ratio*

$$n_{\text{SV}}(\kappa) := 1 + \frac{\lambda_{\max}(\kappa)}{\beta_0}, \quad \beta_0 := \max_{\kappa \in [0, \kappa^*]} (-\lambda_{\max}(\kappa)). \quad (18)$$

The gauge β_0 (the most-stable point of the slow mode) is the SV analogue of the Hawkes baseline rate β at $n = 0$, and is intrinsic to the skeleton (not a tuning parameter).

Theorem 5 (Hawkes-SV equivalence at the Hopf boundary). *1. Critical-endpoint identity. $n_{\text{SV}}(\kappa^*) = 1$, exactly.*

2. Monotonicity past the node-spiral transition. Let $\kappa_{\text{NS}} \in [0, \kappa^)$ denote the smallest κ at which the leading eigenvalue is a complex pair. Then $n_{\text{SV}}(\kappa)$ is monotonically non-decreasing on $[\kappa_{\text{NS}}, \kappa^*]$, strictly increasing wherever $\partial \lambda_{\max} / \partial \kappa > 0$.*

3. Hawkes correspondence. For an exponential-kernel univariate Hawkes process satisfying (3.11), $n_{\text{Hawkes}} = 1 - |\lambda_{\max}^{(\text{Hawkes})}| / \beta$, identical in form to (18). The two coincide at criticality ($n = 1 \Leftrightarrow \lambda_{\max} = 0$), independently of the kernel shape (universality of the stability boundary, Bacry et al., 2015, §2.4).

Proof. Claim (1) is immediate from (18): $\lambda_{\max}(\kappa^*) = 0$ by Hopf-threshold definition, so $n_{\text{SV}}(\kappa^*) = 1 + 0/\beta_0 = 1$. Claim (2): apply the implicit function theorem to the characteristic polynomial at the slow-mode complex pair, which is non-defective on $[\kappa_{\text{NS}}, \kappa^*]$ by definition; Hopf transversality gives $\partial \operatorname{Re} \lambda / \partial \kappa > 0$ at κ^* , propagated by continuity. Claim (3) is the BDHM (2013) diffusive-limit identity restated for the kernel-universal stability boundary via Bacry et al., 2015, §2.4. $\square$

Remark 1 (Honest scope of the equivalence). *Theorem 5 is exact at the criticality endpoint and exact for the 1D exponential-kernel Hawkes globally. It is approximate as a global Hawkes equivalence: Hardiman et al.’s empirical n is the L^1 norm of a fitted multivariate Hawkes kernel on order-flow events, not a direct mapping from the continuous SV state. The identification rests on the universality of the $n = 1 \Leftrightarrow \lambda_{\max} = 0$ boundary across continuous-time reflexive systems, not on a path-by-path identity between event-counting Hawkes processes and our diffusion. We retain κ^* as the natural continuous-time analogue of the Hawkes critical line, and the empirical-phase calibration tests the prediction $n_{\text{SV}}(\kappa_0) \approx 1$ for the SPX-calibrated κ_0 .*

Numerical anchor at the canonical regime. At the section 3.4 regime ($G_x = 0.5$, $G_v = -0.5$, $G_z = -0.5$, $\alpha = 0.5$, $\beta = 1$, $\gamma = 0.5$, $\kappa_v = 2$), (18) evaluated on a 1001-point grid over $\kappa \in [0, 2\kappa^*]$ gives $\beta_0 = 0.214$ attained at $\kappa_{\text{ref}} = 0.124$ (the node-spiral transition, exactly as predicted by claim (2)), $n_{\text{SV}}(\kappa_4^*) = 0.99996$ at the published 4-decimal threshold $\kappa_4^* = 0.8964$, and $n_{\text{SV}}(2\kappa^*) = 2.11$. *Honesty note on the residual:* the $|n - 1| = 3.85 \times 10^{-5}$ at κ_4^* is the truncation error in the published 4-decimal κ^* , not eigenvalue-solver noise — evaluating at the higher-precision Brent root $\kappa^* = 0.8964305216$ returns $|n - 1| < 10^{-15}$, the actual machine- ε floor on a 3×3 matrix; claim (1) of theorem 5 is then a definitional identity ($n_{\text{SV}}(\kappa^*) = 1$ exactly by construction). The published 4-decimal anchor is retained for cross-section consistency with table 1. Figure 8 renders the curve. Implementation: `src/reflexive_options/theory/hawkes_equivalence.py`; reproducer: `python -m reflexive_options.experiments.hawkes_sv_equivalence`.

Empirical position of the SPX market in κ -space. Inverting (18) at the empirical $n_{\text{Hawkes}} = 1.0$ of Hardiman et al. maps directly to $\lambda_{\text{max}} = 0$, i.e., *the empirical SPX market sits exactly at κ^* under any reflexive SV calibration whose deterministic skeleton matches the empirical event-rate ACF in the diffusive limit.* This is the strong testable prediction of theorem 5 for Phase 4: an SPX-calibrated κ_0 should satisfy $n_{\text{SV}}(\kappa_0) \approx 1$. We flag two caveats. First, the Hardiman $n = 1$ is a $\sim 5\%$ -CI band, not a point estimate, so the empirical anchor is the band $n_{\text{Hawkes}} \in [0.95, 1.00]$, equivalently $\kappa \in [\kappa^* - O(0.05/|\partial_\kappa n_{\text{SV}}|), \kappa^*]$. Second, in our *empirical-magnitude* regime ($G_x \sim 10^{-3}$, $\alpha = 252/\text{yr}$) there is no Hopf in the literature-prior range (a finding consistent with section 3.4); the relaxation rate $\beta_0 \approx 2$ is set by the bare Heston-with-memory eigenvalue, and a naive evaluation of n_{SV} at $\kappa_{\text{market}} = 5 \times 10^{-12}$ gives ≈ 0.9998 but this number is dominated by the $\lambda_{\text{max}}(0) = 0$ gauge zero, not by self-excitation. The correct empirical-phase test of the equivalence is therefore: (i) calibrate $G(\cdot)$ from the empirical SPX OI grid (so the G partials inherit the empirical magnitudes and the closed-form κ^* from (14) is the operative threshold); (ii) compute $n_{\text{SV}}(\kappa_0)$ at the calibrated κ_0 ; (iii) compare against the Hardiman band on event-window-matched windows. We pre-commit this protocol to Phase 4.

3.12 Information-theoretic characterization of the critical edge

theorem 1 and theorem 5 characterize κ^* spectrally. We add an *information-theoretic* characterization complementing those spectral statements: how many nats of information does the dealer-gamma channel leak from past spot to future returns at κ^* ?

Setup. Let $y_t = \log(S_t/S^*)$ be the log-deviation of spot, $R_\tau := y_\tau - y_0$ the integrated future log-return, and $\mathcal{F}_{(-\infty, 0]}^y$ the past spot history. Define the *excess entropy at horizon τ* in the ? sense, conditioned on the present non-spot state:

$$E_\tau(\kappa) := I(\mathcal{F}_{(-\infty, 0]}^y; R_\tau \mid v_0, z_0). \quad (19)$$

Conditioning on (v_0, z_0) removes past-spot information already encoded in the present non-spot state, isolating the direct information flow through the dealer-gamma drift $\kappa G(\cdot)$. At $\kappa = 0$ the SDE collapses to standard Heston with Markov closure: future returns are v_0 -conditional independent of past spot, so $E_\tau(0) = 0$. For the linearized 3D OU $dx = J(\kappa)x dt + \Sigma dW$ around the equilibrium with constant-vol surrogate, Gaussian conditioning gives the closed form

$$E_\tau(\kappa) = \frac{1}{2} \log(1 + v_1^2 \cdot \sigma_{y|u,z}^2 / m_y(\tau)), \quad (20)$$

with $v_1 := [e^{J(\kappa)\tau} - I]_{11}$, $\sigma_{y|u,z}^2 := P_{11} - P_{1,(2,3)} P_{(2,3),(2,3)}^{-1} P_{(2,3),1}$, and $m_y(\tau) := [P - e^{J\tau} P e^{J^\top \tau}]_{11}$, where P solves the Lyapunov equation $JP + PJ^\top + \Sigma \Sigma^\top = 0$.

Theorem 6 (Critical excess entropy at the Hopf boundary). *Assume the conditions of theorem 1 hold, so $J(\kappa)$ is Hurwitz on $(\kappa_{\text{NS}}, \kappa^*)$ with a complex pair $\lambda_\pm(\kappa)$ crossing the imaginary axis at κ^* with positive transversality. Let $E_\tau(\kappa)$ be defined by (19) and computed via (20). Then for any fixed $\tau > 0$:*

1. (Markov closure) $\lim_{\kappa \rightarrow 0^+} E_\tau(\kappa) = 0$.
2. (Finite saturation at criticality) $E_\tau(\kappa^*) := \lim_{\kappa \uparrow \kappa^*} E_\tau(\kappa)$ exists, is finite, and is strictly positive.
3. (Mean-field critical exponent $\beta = 1$) $E_\tau(\kappa^*) - E_\tau(\kappa) = C_\tau(\kappa^* - \kappa) + O((\kappa^* - \kappa)^2)$ with $C_\tau > 0$.

4. (*Local monotonicity*) $\exists \delta > 0$ such that $\partial E_\tau / \partial \kappa > 0$ on $(\kappa^* - \delta, \kappa^*)$.

Proof sketch. (1) The first row of $J(0)$ is identically zero in the constant-vol surrogate, so $e^{J(0)\tau}$ has first row $(1, 0, 0)$, giving $v_1 = 0$ and $E_\tau(0) = 0$. (2) As $\kappa \uparrow \kappa^*$ the leading complex pair's real part $\alpha(\kappa) \rightarrow 0$ and $\|P(\kappa)\| \rightarrow \infty$ in the slow-mode eigenvector direction $q = (q_y, q_u, q_z)$. Crucially, this divergence is *coherent* across (y, u, z) — it lives in a 1D subspace. The Schur complement $\sigma_{y|u,z}^2$ extracts the component orthogonal to $\text{span}\{e_2, e_3\}$ in the metric induced by the bounded part of P ; standard low-rank Schur-complement asymptotics (3, §3) give the closed-form limit $\sigma_{y|u,z}^2(\kappa^*) = q_y^2 / (q_u^2 \beta_u + q_z^2 \beta_z)$ for positive constants β_u, β_z from the bounded part of P . Both v_1 and $m_y(\tau)$ stay bounded as $\alpha \rightarrow 0$, so $E_\tau(\kappa^*) < \infty$. (3, 4) Each ingredient in (20) is real-analytic in κ on a punctured neighbourhood of κ^* (?). A first-order Taylor expansion gives the linear scaling with positive coefficient C_τ (the ratio $v_1^2 \sigma^2 / m$ is monotonically increasing in κ on $(\kappa^* - \delta, \kappa^*)$). $\square \square$

Honest scope. The proof gives local monotonicity on a one-sided neighbourhood of κ^* . Global monotonicity on $(0, \kappa^*)$ is checked numerically (below) but not asserted as part of theorem 6 — the Schur-complement asymptotics do not extend across the node-spiral transition κ_{NS} .

Numerical anchor at the canonical regime. `theory/info_theoretic.py` evaluated on a 101-point κ -grid over $(10^{-4}, \kappa^*)$ at the section 3.4 regime (canonical Heston diffusion $\theta_v = 0.04, \xi = 0.3$) returns $E_1(\kappa^*) \approx 0.0530$ nats/yr ($\tau = 1$ yr horizon), $E_{0.1}(\kappa^*) \approx 0.0168$, $E_5(\kappa^*) \approx 0.4135$. Fitted critical exponent $\hat{\beta} = 0.998$ at all three horizons, matching the mean-field prediction $\beta = 1$ to 3 decimals. Global monotonicity holds at every consecutive grid pair across all τ (fig. 9).

Structural insight. The naïve Crutchfield–Feldman intuition would predict $E_\tau(\kappa^*) = \infty$ at criticality; this fails here because the slow-mode collapse is *coherent* across (y, u, z) , and conditioning on (v_0, z_0) projects out the divergent direction. The saturation $E_\tau(\kappa^*)$ is therefore capped by the eigenvector's *spot-purity ratio* $q_y^2 / (q_u^2 + q_z^2)$ at the boundary — a parametric prediction tying qualitative critical behaviour to spot-vs-vol mode partition. A market whose Hopf eigenvector is mostly variance/memory-loaded saturates at a small $E_\tau(\kappa^*)$ even at criticality; a mostly-spot-loaded eigenvector gives a large saturation.

Phase-4 testable prediction. For an SPX market window with calibrated dealer-gamma series $\hat{G}_t$ and observed log-returns $\hat{r}_{t+1}$, the empirical ? transfer entropy $\hat{T}_{G \rightarrow r} := H(\widehat{r_{t+1}} | r_t) - H(r_{t+1} | \widehat{r_t}, G_t)$ should be statistically significant ($p < 0.05$) under an IAAFT-surrogate null on the source series AND should be larger on event windows where the system is conjectured to sit closer to κ^* (Volmageddon, COVID, Yen carry) than on quiescent windows. This corollary is logically independent of the theorem 5 prediction $n_{\text{SV}}(\kappa_0) \approx 1$: theorem 5 measures *how close* the system is to the critical edge; theorem 6 measures *how much past-spot information* the system actually leaks at that distance. Implementation: `transfer_entropy_iaaft_pvalue` in `theory/info_theoretic.py`; the IAAFT-null convention matches pre-registration amendment A5. Relation to ?'s active-information-storage framework: the conditioning on (v_0, z_0) in (19) is precisely their fixed-history-baseline construction, applied to a continuous-time SDE channel at the Hopf boundary.

4 Numerical phase diagram

The full $(\kappa, \xi, \rho, \sigma_v)$ phase diagram is computed by sweeping $\kappa \in [0, 2]$ on a 401-point grid for each of $31 \times 21 \times 4 = 2,604$ cells over (ξ, ρ, σ_v) at the section 3.4 Hopf-exhibiting regime. The pair (ξ, ρ) enters the deterministic Jacobian via the leading shear-induced correction $a_{\text{eff}}(\kappa) = a(\kappa) + \frac{1}{2}\xi^2 \rho G_v$ —an

Engel et al. (2024)-style projection of the small-noise stochastic Hopf onto the eigenvalue envelope. Wall-clock: ~ 16 s on a single M-series core. Figure 10 renders the (ξ, ρ) heatmap and four line cuts of $\kappa^*(\sigma_v)$ at SPX-relevant (ξ, ρ) probes.

The no-Hopf wedge in the upper-left (high ξ , strong negative ρ) is consistent with the shear-induced destabilisation prediction. The four line cuts substantiate the section 3.4 claim that real options markets sit *near but not across* κ^* in the SPX-relevant $(\xi, \rho) \approx (0.30, -0.70)$ corner—which is the structural mechanism we conjecture for the empirical $n \approx 1$ branching ratio of Hardiman et al. (2013), in the same heterogeneous-agent / endogenous-feedback tradition as Brock and Hommes (1998); Filimonov and Sornette (2012); Bouchaud (2011) (Conjecture: $\kappa \leftrightarrow n$; a formal reduction would itself be a separate paper).

Figure 11 renders the (σ_q, γ) phase boundary from the closed-form ℓ_1 of section 3.5. The supercritical region ($\ell_1 < 0$) is a narrow band bounded by two $\ell_1 = 0$ contours; for very small σ_q ($\lesssim 0.05$) the band collapses entirely. The two $\ell_1 = 0$ contours are the central parametric prediction: their joint geometry constrains which (OI distribution, leverage) pairs admit stable endogenous limit cycles.

5 Pre-registered evaluation framework

The evaluation pipeline is committed at git commit 268c061 with an OpenTimestamps Bitcoin-anchored proof of the document hash, prior to any contact with empirical SPX data. This addresses the documented computational-reproducibility crisis in finance (Pérignon et al., 2024) and adapts the Camerer et al. (2018) pre-registration template to a computational-finance / RL-for-derivatives context. We do not claim “first pre-registration in finance”—the Pacific-Basin Finance Journal pathway (Faff, 2022) and the AEA RCT Registry (American Economic Association, 2018) predate this work. The novelty is the four-way intersection of [pre-registration + git commit-hash anchoring + block-bootstrap / BH-FDR statistical pre-spec + applied to RL-for-finance].

Two evaluation primitives. (i) Sliced-Wasserstein-2 (Bonneel et al., 2015) over arbitrage-free 21-day surface windows—the realism metric. (ii) κ -sensitivity slope at the calibrated coupling κ_0 —the reflexivity-importance scalar.

H1 (primary realism). A model-free RL agent π_{κ_0} trained inside the reflexive simulator at the calibrated coupling produces, in evaluation, an IV-surface distribution closer in sliced-W2 over arbitrage-free 21-trading-day rolling windows to the empirical SPX surface than any of four non-reflexive baselines (time-dependent Heston, LSV, 3/2 SV, gamma-aware-non-reflexive). Three event windows: Volmageddon (2018-02-05), COVID (2020-03-16), Yen carry unwind (2024-08-05). Decision rule: 12 pairwise dominance checks (4 baselines $\times$ 3 windows) with non-overlapping 95% block-bootstrap CIs (Politis and Romano, 1994). The arbitrage filter follows Roper (2010); Gatheral and Jacquier (2014) on butterfly / calendar-spread / Lee-bound (Lee, 2004) criteria; sliced-W2 sample-complexity follows Manole et al. (2022).

Sliced-W2 sample-complexity audit. A pre-data audit (`scripts/sw2_sample_complexity.py`, output at `runs/sw2_sample_complexity/`) characterises the bootstrap-95%-CI half-width on sliced-W2 between two Heston populations differing only in their long-run variance ($\theta_A = 0.04$ vs $\theta_B = 0.0576$, equivalently σ_{LR} from 20% to 24%). Each side draws from a 5,000-day population; the “true” SW2 is averaged over 10 slice-direction MC repeats on the full populations and equals $\widehat{\text{SW2}}_{\text{true}} = 0.0555$. For $n_{\text{windows}} \in \{30, 100, 300, 1,000, 3,000\}$ we sub-sample n_{windows} rolling 21-day

windows from each side, compute the SW2 estimate, and report the median bootstrap CI half-width across 30 outer seeds (table 4, fig. 12).

n_{windows}	median SW2	median CI half-width	ratio to SW2 _{true}
30	0.0611	0.0300	0.541
100	0.0545	0.0195	0.352
300	0.0548	0.0124	0.223
1,000	0.0534	0.0085	0.153
3,000	0.0553	0.0065	0.118

Table 4: Sliced-W2 sample-complexity at the Heston-A vs Heston-B contrast. The half-width follows a power law $\text{ratio}(n) \approx 1.64 \cdot n^{-0.337}$ ($R^2 \geq 0.99$), giving an extrapolated $n_{\min} \approx 4,000$ windows for the $\pm 10\%$ half-width target. Run dir: `runs/sw2_sample_complexity/`.

H1 power gating. Each H1 event window contributes ~ 280 rolling 21-day windows (300 trading days $\times$ stride 1, minus the 20-day burn-in). The H1 budget per event is therefore ~ 280 windows, well below the $\sim 4,000$ extrapolated $n_{\min}$ for an absolute $\pm 10\%$ half-width on sliced-W2. The *decision rule* (non-overlapping 95% CIs between distinct baselines) does not require the absolute $\pm 10\%$ precision target—it requires the inter-model SW2 gap to exceed twice the within-model half-width. At $n = 280$, the in-grid linear interpolation of table 4 gives a half-width / SW2_{true} ratio of ≈ 0.23 , so two distinct H1 baselines must differ in SW2 by at least $\approx 0.46 \cdot \text{SW2}_{\text{true}}$ to be discriminable at 95% confidence in a single event. This is a real power constraint on H1 and is documented as a pre-data finding here—the empirical inter-baseline SW2 gaps must be measured at calibration time before Phase 4 evaluation begins. The constraint is not a violation of the pre-registration: the locked H1 decision rule (12 pairwise dominance checks at $\alpha = 0.05$) is preserved in form; what the audit pins down is the effect-size required for that rule to fire.

H2–H4 (secondary). H2 is the κ -sensitivity slope $\tilde{\rho} = \partial \hat{m} / \partial \kappa|_{\kappa_0}$ —positive and statistically distinguishable from zero for the reflexive-trained agent, statistically indistinguishable for baselines (TOST equivalence, applied to the dimensionless elasticity per A7, with the GP slope CI computed under the pre-registered RBF + WhiteKernel kernel of A6). *Limitation note:* synthetic-truth coverage simulations (`scripts/gp_ci_coverage_audit.py`) at the H2 evaluation budget show the nominal-95% RBF-GP slope CI at empirical 91.5–100% coverage on smooth-truth functions ($\sigma = 0.10$ noise, single-seed condition) but drops to 67–94% under tighter $\sigma = 0.05$ multi-seed conditions; an exploratory Matérn-3/2 alternative was attempted but the derivative posterior of a once-differentiable kernel produces non-shrinking finite-difference variance and yields vacuous CIs. Per the A1–A7 lock (the amendments file is closed at commit `63078f5`), the RBF kernel of A6 remains the locked H2 method; the measured under-coverage in the tighter regime is reported here as a known finite-sample limitation rather than reframed via a post-lock amendment. H3 is the post-FOMC RR25 and ATM-term-structure shifts. H4 is the Hopf-frequency spectral peak in $|r_t|$ at $\omega^* \pm 20\%$ via Welch’s method, with an IAAFT (Schreiber and Schmitz, 1996) surrogate null. The H4 detector is implemented and unit-tested against the locked decision rule (in-band $\wedge p < \alpha/m$) in `tests/test_spectral_amendments.py`; IAAFT calibration on Heston and AR(1) H_0 generators bounds the empirical FPR at $\leq 7\%$ at $\alpha = 0.05$.

H4 detector power on realistic positive controls. `scripts/h4_power_realistic.py` (output at `runs/h4_power_realistic/`) characterises detector power against two realistic positive controls: (i) the Stuart–Landau oscillator $\dot{z} = (\mu + i\omega^*)z - |z|^2z + \sigma dW$ at $\omega^* = 1$ cyc/yr for $\mu \in \{0.1, 0.5, 1.0\}$ and $\sigma \in \{0.05, 0.10, 0.20\}$, and (ii) the supercritical reflexive simulator (section 3.4 regime, trivial-OI configuration) at $\kappa \in \{1.05, 1.20, 1.50\} \cdot \kappa^*$. Each cell uses 50 seeds and 75 IAAFT permutations; fig. 13 reports power vs trajectory length $T \in \{256, 512, 1024, 2048\}$.

The headline result: **the H4 detector achieves $\geq 80\%$ peak power against the Stuart–Landau positive control *specifically* at $T = 512$ daily samples** (~ 2 years of trading days at $f_s = 252/\text{yr}$) for 8/9 tested (μ, σ) configurations under the locked IAAFT-surrogate $\alpha = 0.05$ rule. Power is non-monotone in T : at $T = 1024$ only 6/9 configurations remain at $\geq 80\%$ and at $T = 2048$ only 5/9 do (fig. 13). This degradation is the IAAFT-conservativeness signature acting in the opposite direction — as T grows the surrogate’s preserved linear ACF approximates the underlying limit cycle more faithfully, narrowing the p -value gap. The implication for empirical Phase 4 is that detector power is maximised at the $T \approx 500$ – 1000 sample range matching our locked event windows, not at indefinite-horizon trajectories. This is nevertheless a substantively stronger claim than the v0.3.1 deferral and provides direct evidence that the detector is functional against realistic limit-cycle data—Stuart–Landau (Kuznetsov, 2004) being the canonical Hopf normal form to which the supercritical section 3.4 skeleton reduces by Hartman–Grobman near κ^* . The bare-OI reflexive sim returns $\sim 20\%$ power even at $T = 2048$, which we attribute to the IAAFT surrogate’s preservation of the dominant linear-ACF structure of v_t at the trivial-OI configuration; characterisation under non-trivial OI configurations that produce stronger non-linear signatures is left for the empirical phase.

Differentiation from precedents. Ning et al. (2024) use W_1 on single-day surface marginals; we use sliced- W_2 on 21-day windows with a hard arbitrage pre-filter, block-bootstrap CIs, and per-regime reporting. He et al. (2025) (NeurIPS 2025) sweep a Wasserstein-ball perturbation budget with a separate adversarially-trained agent at each grid point; we deploy a single agent across the κ grid and report the slope at anchor with a GP-posterior CI (Ma and Huang, 2025) for the elliptic-uncertainty-set analogue. Subbaswamy et al. (2022) (NeurIPS 2022) provide the closest theoretical precursor for slope-at-anchor robustness summaries (parametric robustness sets, local 2nd-order approximation), which we operationalise empirically and apply to RL-for-finance (Bai et al., 2024; Huang et al., 2024). Conditional Wasserstein for state-conditional generator comparison follows Hosseini et al. (2023). Building blocks: Vuletić and Cont (2025); Choudhary et al. (2024); Issa et al. (2023); Jin and Agarwal (2025); Chevallier et al. (2025); Luan and Hamp (2025); Buehler et al. (2019); Cao et al. (2021); Packer et al. (2018); Murray et al. (2022).

5.1 Synthetic pipeline validation

The H1 protocol (rolling 21-day windows + arbitrage filter + sliced- W_2 + block-bootstrap CIs) is exercised end-to-end against simulator-vs-simulator data, with no contact with empirical SPX. Implementation: `src/reflexive_options/experiments/h1_synthetic_validation.py`; tests in `tests/test_h1_synthetic_validation.py`. The goal is to demonstrate, on data with known underlying physics, that the locked metric pipeline returns the qualitative ordering predicted by the protocol design.

Three-source synthetic ground truth. We train a small BC-MLP agent π_{κ_0} inside the reflexive simulator at a synthetic-validation anchor $\kappa_0 = 10^{-10}$ (set above the literature prior so the κ -effect on dynamics is detectable at the path budget — the empirical pipeline uses the calibrated empirical

κ_0 , not this synthetic anchor). Each rollout collects a 100-day per-day surface trajectory. Three sources, each yielding $n = 30$ trajectories at the production budget: (a) κ_0 *deployed* — the trained agent acts inside the SAME reflexive simulator at κ_0 (different seed sequence from the reference κ_0 -training distribution; expected to be statistically indistinguishable from the reference); (b) $2 \times \kappa_0$ *reflexive* — the trained agent acts inside the reflexive simulator at $4 \cdot \kappa_0$ (a four-fold coupling change at the synthetic anchor; expected to be measurably further from the reference); (c) *Heston (different mechanism)* — forward-simulated Heston with no dealer-gamma coupling at all (different mechanism class; expected to be even further). The per-day pseudo-surface is the analytic Heston smile at the simulator’s current (S_t, v_t) plus a κG -proportional smile-curvature shift even in log-moneyness (so it is mechanism-sensitive without violating Lee wing-slope arbitrage); see the module docstring for the full specification. The Heston source uses the more aggressive $(\rho, \xi) = (-0.70, 0.30)$ backbone vs the synthetic reflexive sources’ tame $(-0.30, 0.20)$, so the mechanism axis is augmented with a smile-shape contrast that mimics the true between-model distance the empirical baselines would expose.

Result. On the production-budget run (BC train: 30 episodes; per-source: 30 rollouts $\times$ 100 days; 200 stationary block-bootstrap resamples), the sliced- W_2 ordering and the locked 95% CI separation are:

Target source	SW2 vs κ_0 training distribution	95% CI low	95% CI high
(a) κ_0 deployed	0.00498	0.00431	0.0164
(b) $2 \times \kappa_0$ reflexive	0.0337	0.0300	0.0402
(c) Heston (mechanism)	0.0541	0.0434	0.0611

Table 5: H1 synthetic-validation sliced- W_2 ordering. The expected ordering $\text{SW2(a)} < \text{SW2(b)} < \text{SW2(c)}$ holds; the 95% block-bootstrap CIs of (a) vs (b) and (b) vs (c) are both non-overlapping. All three pre-registered acceptance conditions for the synthetic-validation protocol are satisfied. Run dir: `runs/h1_synthetic/`; figure: fig. 14.

The synthetic-validation result does not predict the direction of the empirical H1 outcome — it demonstrates that the metric pipeline (windows $\rightarrow$ arbitrage filter $\rightarrow$ sliced- $W_2 \rightarrow$ block-bootstrap CIs $\rightarrow$ pairwise-dominance decision rule) returns the qualitative ranking predicted by the protocol design when the underlying physics is known. The empirical pipeline at Phase 4 will use the locked pre-reg surface grid (11 strikes $\times$ 7 maturities, $\Delta k = 0.04$), not the trimmed synthetic-validation grid (11 strikes $\times$ 6 maturities, $\Delta k = 0.03$); the difference is documented in the experiment module and is necessary because the QuantLib analytic Heston engine returns NaN at the deep wings for the shortest pre-reg maturities at the synthetic-anchor variance level. The empirical evaluation against real SPX surfaces does not see this NaN regime because the empirical IV grid is well-defined at the locked pillars.

6 Mechanism decomposition vs Marketron

The reflexive simulator and the Marketron quasi-particle SDE (Halperin and Itkin, 2025) are not the same mechanism, and a 1:1 reproduction of Marketron’s Tables 7–8 from our simulator is mathematically impossible: in our model the memory carrier is the low-pass-filtered log-price z entering only the variance drift via γz ; in Marketron it is the unobservable y entering both drifts

non-linearly through a potential $V_M(x)$. Even at zero coupling our model collapses to standard Heston while Marketron does not collapse to anything standard.

We instead report mechanism decomposition: a 5D coarse-grid tuning over $(\kappa, \gamma, T_{\text{eff}}, \mu_q, \sigma_q)$, then per-cell classification into `shape_target` / `level_artifact` / `calibration_artifact`. The headline gate is the shape-match rate on `shape_target` cells.

Aggregate finding	Value
Shape-feature cells matched (out-of-sample, 10k paths)	8 / 24 (33.3%)
$\geq 30\%$ gate (one-sided binomial $p \approx 0.27$)	passed
Long-horizon (≥ 0.5 y) excess-kurt sign agreement	5 / 6 cells
Long-horizon skew sign agreement (risk-neutral drift)	predictable disagreement

Table 6: Mechanism-decomposition headline. Long-horizon excess-kurt sign agrees with Marketron Table 8 on 5/6 horizons; long-horizon skew sign disagrees predictably under risk-neutral drift (Marketron’s positive long-horizon skew comes from Bessembinder (2018) compounding under calibrated drift; ours comes from dealer-gamma + leverage feedback under $\mu = 0$, see also Farago and Hjalmarrsson, 2023).

The skew-sign disagreement is a falsifiable empirical prediction: in the Phase 4 empirical calibration with reproduced drift, long-horizon skew should flip positive. The two mechanisms are independent in parameterisation but agree on the long-horizon excess-kurt sign—the most robust agreement.

6.1 A priori mechanism-relevant cell restriction

The 8/24 (33.3%) headline counts *all* shape-target cells across all horizons — a deliberately conservative aggregate that carries no a priori physical structure. The dealer-gamma feedback channel does not predict sign agreement on every shape cell uniformly; it predicts agreement on a specific subset where the channel has time to integrate and where the comparison is not contaminated by mechanism-extraneous calibration choices. We pre-commit (via the `is_mechanism_relevant_cell` predicate in `src/reflexive_options/experiments/synthetic_replication.py`, locked before per-cell outcomes are inspected) to the following a priori restriction:

- *Long-horizon* ($T \geq 0.5$ yr) shape cells only — short horizons are dominated by the variance-OU mean-reversion and do not yet reflect the dealer-gamma signature.
- *Skew + excess-kurt* only (the `shape_target` class) — mean is dominated by Marketron’s calibrated $\bar{\eta} + f(\theta) - c(t)V'_M(x)y$ drift terms we do not model (`calibration_artifact`) and vol is documented in the Marketron brief §6.4 as 3–5× too high vs realised SPX (`level_artifact`).
- *Within-envelope* cells only — the high- σ Marketron Table 6 calibration sits at the simulator’s variance-truncation envelope at the per-set tuned coupling (cf. `paper/mechanism_decomposition.md` Table 2 and section 3). Cells with $|\hat{m}| \geq 10$ (above the Marketron-published shape-moment range by an order of magnitude) or NaN measurements are flagged *envelope-excluded* a priori, akin to instrument saturation, and dropped from the binomial denominator. The envelope bound and predicate are constants of the source code (`SHAPE_ENVELOPE_ABS_BOUND`, `LONG_HORIZON_THRESHOLD_YEARS`) committed before per-cell outcomes were inspected.

Restricted result. At the in-sample budget (n_paths=2000 used by the tuning sweep), the a priori-restricted subset has 7 matches out of 10 finite-and-in-envelope mechanism-relevant cells (rate = 70%, one-sided binomial $p = 0.172$ under the chance-agreement null). At the out-of-sample budget (n_paths=10000), the restricted subset has 4/8 matches (rate = 50%, $p = 0.637$), driven primarily by Table 5 long-horizon skew flipping more strongly negative at the larger sample (consistent with the documented risk-neutral-drift skew disagreement in table 6). *The strengthened restriction does not reach $p < 0.05$ at either budget given the small subset cardinality*, and we therefore retain the all-cell 8/24 (33.3%) as the headline number with the restricted result reported as a transparent secondary statistic. The restriction *is* the right physical aggregator for the dealer-gamma mechanism — but the present sub-sample size at the per-set tuned coupling is too small to lift the chance-null rejection below $p = 0.05$, and the empirical Phase 4 evaluation (which will include far more horizons-by-event windows) will be needed to adjudicate. Implementation: `aggregate_mechanism_relevant_subset(...)` in `src/reflexive_options/experiments/synthetic_replication.py`; pre-data anchored test in `tests/test_markettron_tuning.py::test_mechanism_relevant_subset_match_rate_exceeds_chance_threshold`.

7 Conclusions and future work

We have introduced a 3D reflexive SV simulator (S_t, v_t, z_t) with explicit dealer-gamma drift, proved a Hopf bifurcation theorem with a closed-form first Lyapunov coefficient ℓ_1 and Hopf threshold κ^* for log-normal open-interest in moneyness, computed a numerical (ξ, ρ, σ_v) phase diagram and a (σ_q, γ) closed-form phase boundary, and pre-registered the empirical evaluation pipeline (sliced- W_2 , κ -sensitivity slope, Hopf-frequency spectral test) at a verifiable git commit hash with an OpenTimestamps Bitcoin-anchored proof.

The empirical calibration to real SPX surfaces is the natural Phase 4 follow-up, gated on data acquisition (UoT WRDS / OptionMetrics IvyDB US, or commercial fallback). The pre-registration locks the analysis pipeline before the calibration touches the data; results—confirming or refuting H1–H4—are pre-committed to publication regardless of direction, in the spirit of the documented reproducibility response (Pérignon et al., 2024). Theorem 5 (section 3.11) closes the long-standing gap between the discrete-time Hawkes literature and continuous-time SV bifurcation analysis at the criticality endpoint, identifying the empirical Hardiman et al. (2013) $n \approx 1$ with our κ^* via the ? diffusive-limit identity and the kernel-universal stability boundary of Bacry et al. (2015). Remaining open theoretical items: closed-form $\Lambda(\kappa)$ via Engel–Lamb–Rasmussen-style asymptotics adapted to Heston multiplicative noise (currently numerical only); extending theorem 5 from the criticality-endpoint identity to a global path-by-path equivalence between event-counting Hawkes processes and our continuous-state diffusion (cf. Bondi et al., 2024).

The Fokker–Planck stationary marginal density of the reflexive simulator deviates from the closed-form Heston comparator (Feller, 1951) on tail index and skew sign in a way that tracks $\text{sgn}(G_x)$. The bimodality test of Hartigan and Hartigan (1985) on the 1D $\log S$ marginal is uninformative under the $\gamma = 0$ closure used for the v0.3.0 scan ($p \geq 0.10$ across the κ -grid); a follow-up scan with $\gamma > 0$ active and the dip statistic computed on the principal-component projection of the 2D $(\log S, v)$ joint density flips to *supported* ($p = 0.033$) at $\kappa = 1.05 \kappa_{\text{env}}^*$ (fig. 15), where $\kappa_{\text{env}}^* \approx 3.9 \times 10^{-9}$ is the simulator’s stability-envelope upper bound (distinct from the deterministic Hopf threshold $\kappa^* \approx 0.896$ of section 3.4). The dip statistic is computed on the $n = 15,769$ surviving cells out of 20,000 post-burn-in (path $\times$ kept-step) entries ($\sim 79\%$ survival at this κ in the canonical regime); the structural caveat is that the surviving subsample is conditioned on path non-blowup, which itself shapes the joint distribution—we therefore report the result as preliminary rather than definitive.

Detector validation uses the Benettin et al. (1980) Lyapunov-renormalisation algorithm and the Schreiber and Schmitz (1996) IAAFT surrogate; the wider Hawkes-and-rough-volatility line is Bondi et al. (2024); Bacry et al. (2015); Livieri et al. (2018); arbitrage-free neural SDE market models follow Cohen et al. (2023).

All experiments are reproducible via `bash scripts/verify.sh` from commit 268c061; pre-registration anchored via OpenTimestamps. Code: github.com/mahimn01/reflexive-options.

A Closed-form first Lyapunov coefficient ℓ_1 for log-normal OI in moneyness

This appendix renders the layered closed-form expression for $\ell_1(\kappa^*)$ (Eq. 15) referenced in section 3.5. The bare expanded form is a ≈ 7.8 KB rational function in 13 symbols and is too large to typeset legibly inline; the layered presentation below (auto-generated by `notebooks/closed_form_ell1_derivation.py` and verified numerically against `lyapunov_coefficient_lognormal_o1` to $\sim 10^{-13}$ relative at the canonical regime) is the practical statement.

Layer 1 — Hopf eigenvectors (right and left). At $\kappa = \kappa^*$, the right eigenvector q of J for the imaginary eigenvalue $+i\omega^*$, parametrised by $q_3 = 1$, is

$$q_1 = \frac{\gamma(-2G_v\kappa + 1)}{2(G_y\kappa\kappa_v + iG_y\kappa\omega - i\kappa_v\omega + \omega^2)}, \quad (21)$$

$$q_2 = \frac{\gamma}{\kappa_v + i\omega}, \quad (22)$$

$$q_3 = 1. \quad (23)$$

The left eigenvector $\tilde{p}$ of $J^\top$ for $-i\omega^*$, parametrised by $\tilde{p}_3 = 1$, is

$$\tilde{p}_1 = -\frac{\beta}{G_y\kappa + i\omega}, \quad (24)$$

$$\tilde{p}_2 = \frac{\beta(-2G_v\kappa + 1)}{2(G_y\kappa\kappa_v - iG_y\kappa\omega + i\kappa_v\omega + \omega^2)}, \quad (25)$$

$$\tilde{p}_3 = 1. \quad (26)$$

Imposing the bilinear normalisation $\langle p, q \rangle := \sum_i \bar{p}_i q_i = 1$ rescales $p = \tilde{p}/\bar{N}$ with

$$N := \langle \tilde{p}, q \rangle = \frac{\beta\gamma(\kappa_v + i\omega)(2G_v\kappa - 1) - \beta\gamma(2G_v\kappa - 1)(G_y\kappa - i\omega) + 2(\kappa_v + i\omega)(G_y\kappa - i\omega)(G_y\kappa\kappa_v + iG_y\kappa\omega - i\kappa_v\omega + \omega^2)}{2(\kappa_v + i\omega)(G_y\kappa - i\omega)(G_y\kappa\kappa_v + iG_y\kappa\omega - i\kappa_v\omega + \omega^2)} \quad (27)$$

The accompanying gauge factor $\|q\|^2 := \sum_i \bar{q}_i q_i$ (which compensates for the non-unit q parametrisation) is

$$\|q\|^2 = \frac{G_v^2\gamma^2\kappa^2 - G_v\gamma^2\kappa + G_y^2\gamma^2\kappa^2 + G_y^2\kappa^2\kappa_v^2 + G_y^2\kappa^2\omega^2 + \gamma^2\omega^2 + \frac{\gamma^2}{4} + \kappa_v^2\omega^2 + \omega^4}{G_y^2\kappa^2\kappa_v^2 + G_y^2\kappa^2\omega^2 + \kappa_v^2\omega^2 + \omega^4}. \quad (28)$$

Substituting $\omega^{*2} = c_1(\kappa^*) = -G_y\kappa(\alpha + \kappa_v) + \alpha\kappa_v$ into the above eliminates ω^* in favour of $(\kappa^*, G_y, \kappa_v, \alpha)$.

Layer 2 — Bautin contraction terms. Define the three Kuznetsov 2004 eq. 3.20 contractions

$$\mathcal{T}_1 := \langle p, C(q, q, \bar{q}) \rangle, \quad (29)$$

$$\mathcal{T}_2 := -2 \langle p, B(q, J^{-1}B(q, \bar{q})) \rangle, \quad (30)$$

$$\mathcal{T}_3 := \langle p, B(\bar{q}, (2i\omega^*I - J)^{-1}B(q, q)) \rangle. \quad (31)$$

Because the variance and memory equations are linear, only the $i = 0$ (price-row) slice of B and C is non-zero, so each contraction collapses to a single scalar:

$$B(u, v)_1 = \kappa^* [G_{aa}u_1v_1 + G_{av}(u_1v_2 + u_2v_1) + G_{vv}u_2v_2], \quad (32)$$

$$\begin{aligned} C(u, v, w)_1 = \kappa^* [G_{aaa}u_1v_1w_1 + G_{aav}\sum_{\sigma} u_{\sigma(1)}v_{\sigma(2)}w_{\sigma(3)} \\ + G_{avv}\sum_{\sigma} u_{\sigma(1)}v_{\sigma(2)}w_{\sigma(3)} + G_{vvv}u_2v_2w_2], \end{aligned} \quad (33)$$

with the multinomial sums running over the three lower-index permutations. Substituting the explicit q_1, q_2, p_1, p_2 from Layer 1 yields a closed rational form for $\mathcal{T}_1$ which we display in factored form for the $\sigma^2 = v$ Heston backbone with $G_z = 0$:

$$\mathcal{T}_1 = \frac{1}{N} \left(\frac{-8G_{aaa}G_v^3\beta\gamma^3\kappa^4 + 12G_{aaa}G_v^2\beta\gamma^3\kappa^3 - 6}{8G_vG_y^3\beta\gamma\kappa^4\kappa_v - 8iG_vG_y^3\beta\gamma\kappa^4\omega - 8G_vG_y^2\beta\gamma\kappa^3\kappa_v^2 - 8iG_vG_y^2\beta\gamma\kappa^3\kappa_v\omega - 16G_vG_y^2\beta\gamma\kappa^3\omega^2 + 8G_vG_y\beta\gamma\kappa^2\kappa_v\omega} \right) \quad (34)$$

(Here we have factored out $1/N$ from the $\bar{p}$ normalisation; the remaining bracket is a polynomial in the G_* partials and the rational in $(\kappa^*, \omega^*, \kappa_v, \alpha, \beta, \gamma)$ implied by the Layer 1 q_i, p_i . The corresponding $\mathcal{T}_2, \mathcal{T}_3$ acquire denominator $\det J$ and $\det(2i\omega^*I - J)$ respectively, so they are inconveniently long to typeset inline; their construction is the same.)

Layer 3 — Closed-form ℓ_1 (Eq. 19). The first Lyapunov coefficient is the gauge-corrected real part:

$$\ell_1(\kappa^*) = \frac{1}{2\omega^* \|q\|^2} \text{Re}[\mathcal{T}_1 + \mathcal{T}_2 + \mathcal{T}_3]. \quad (19)$$

Term-power scaling and limits. Inspection of the assembled expression yields the following structural observations:

- Each contraction $\mathcal{T}_k$ carries a factor of κ^{*3} (one κ^* per non-linear vertex; C -vertices in $\mathcal{T}_1$ give a single κ^* , while $\mathcal{T}_{2,3}$ contain two B -vertices joined by a resolvent — each B contributes one κ^*). After dividing by $2\omega^*\|q\|^2$, the leading κ^* -power of ℓ_1 is κ^{*2} when $\|q\|^2$ is dominated by its constant term and degenerates to κ^{*0} in the opposite limit.
- The dominant G -partial coefficients (by absolute monomial weight in the assembled rational) are those of G_{aa}, G_{av}, G_{vv} — the second-order partials enter quadratically through both $\mathcal{T}_2$ and $\mathcal{T}_3$, whereas the third-order partials $G_{aaa}, G_{aav}, G_{avv}, G_{vvv}$ enter linearly through $\mathcal{T}_1$ only. Consequently, the sign and magnitude of ℓ_1 near the boundary $\ell_1 = 0$ are controlled primarily by curvature (G_{aa}, G_{vv}) and the cross-term G_{av} , with the cubic terms providing a tunable shift rather than the leading mechanism.
- *Limit $\gamma \rightarrow 0$:* the leverage channel switches off, $L = \beta\gamma \rightarrow 0$, and the constant term of $H(\kappa)$ (Eq. 13) becomes $\alpha\kappa_v(\alpha + \kappa_v) > 0$, so $H(0) > 0$ and the smallest positive root of the quadratic

(when one exists) drifts to $\kappa^* \rightarrow (\alpha + \kappa_v)/G_y$ from below; ℓ_1 inherits this κ^* -dependence directly. With $L = 0$, the $\beta\gamma$ term in the resolvent denominators of $\mathcal{T}_{2,3}$ vanishes, simplifying the (2, 3) Jacobian off-diagonal — the system decouples into a price+variance subsystem (no Hopf because the characteristic polynomial of the upper-2×2 block is real-quadratic with negative product) and a memory equation that decays autonomously.

- *Limit $\alpha \rightarrow \infty$* : the memory channel becomes fast (z relaxes on a vanishing timescale), $A = \alpha + \kappa_v \rightarrow \infty$, and $\kappa^* = (G_y A^2 - G_v L - \sqrt{D})/(2G_y^2 A) \sim A/G_y \rightarrow \infty$ — the Hopf threshold is pushed to operationally inaccessible coupling. ℓ_1 then carries a $\kappa^{*2} \sim \alpha^2$ enhancement from the $\mathcal{T}_k$ vertices, but is divided by $2\omega^* \|q\|^2$ which itself grows as $\sqrt{\alpha\kappa_v} \cdot \alpha^2$ in the leading limit; the net ratio is $\ell_1 \sim \sqrt{\alpha}$, i.e. supercritical Hopf becomes *stronger* in the fast-memory limit.
- *Limit $\kappa_v \rightarrow \infty$ (fast variance mean-reversion)*: the Heston backbone collapses to constant variance and $b(\kappa^*) = \kappa^* G_v - 1/2 \rightarrow -1/2$ at fixed κ^* , so the Hopf mechanism reduces to the price↔memory feedback loop only. The closed-form κ^* shifts as $A^{-1} \rightarrow 0$ but the leading balance of ℓ_1 tracks the same $\sqrt{\kappa_v}$ scaling as the α limit by symmetry of the formulas under (α, κ_v) swap-with-sign-conventions on $b(\kappa^*)$.

References

- American Economic Association. AEA RCT registry. <https://www.aeaweb.org/journals/policies/rct-registry>, 2018.
- Ludwig Arnold. *Random Dynamical Systems*. Springer, 1998.
- Emmanuel Bacry, Iacopo Mastromatteo, and Jean-François Muzy. Hawkes processes in finance. *Market Microstructure and Liquidity*, 1(1), 2015.
- Yahui Bai, Yuhe Gao, Runzhe Wan, Sheng Zhang, and Rui Song. A review of reinforcement learning in financial applications. *arXiv preprint arXiv:2411.12746*, 2024. URL <https://arxiv.org/abs/2411.12746>.
- Andrea Barbon and Andrea Buraschi. Gamma fragility. *SSRN Working Paper*, 2021.
- Peter H. Baxendale. A stochastic Hopf bifurcation. *Probability Theory and Related Fields*, 99: 581–616, 1994.
- Giancarlo Benettin, Luigi Galgani, Antonio Giorgilli, and Jean-Marie Strelcyn. Lyapunov characteristic exponents for smooth dynamical systems and for Hamiltonian systems. *Meccanica*, 15:9–30, 1980.
- Hendrik Bessembinder. Do stocks outperform treasury bills? *Journal of Financial Economics*, 129(3):440–457, 2018. doi: 10.1016/j.jfineco.2018.06.004.
- Alessandro Bondi, Sergio Pulido, and Simone Scotti. The rough Hawkes Heston stochastic volatility model. *Mathematical Finance*, 34(4), 2024.
- Nicolas Bonneel, Julien Rabin, Gabriel Peyré, and Hanspeter Pfister. Sliced and Radon Wasserstein barycenters of measures. *Journal of Mathematical Imaging and Vision*, 51(1):22–45, 2015. doi: 10.1007/s10851-014-0506-3.

- Jean-Philippe Bouchaud. The endogenous dynamics of markets: Price impact, feedback loops and instabilities. In *Lessons from the Financial Crisis*. Risk Books, 2011.
- William A. Brock and Cars H. Hommes. Heterogeneous beliefs and routes to chaos in a simple asset pricing model. *Journal of Economic Dynamics and Control*, 22(8–9):1235–1274, 1998. doi: 10.1016/S0165-1889(98)00011-6.
- William A. Brock, Cars H. Hommes, and Florian O. O. Wagener. More hedging instruments may destabilize markets. *Journal of Economic Dynamics and Control*, 33(11):1912–1928, 2009. doi: 10.1016/j.jedc.2009.05.004.
- Hans Buehler, Lukas Gonon, Josef Teichmann, and Ben Wood. Deep hedging. *Quantitative Finance*, 19(8):1271–1291, 2019. doi: 10.1080/14697688.2019.1571683.
- Colin F. Camerer, Anna Dreber, Felix Holzmeister, Teck-Hua Ho, Jürgen Huber, Magnus Johannesson, Michael Kirchler, Gideon Nave, Brian A. Nosek, Thomas Pfeiffer, et al. Evaluating the replicability of social science experiments in Nature and Science between 2010 and 2015. *Nature Human Behaviour*, 2:637–644, 2018. doi: 10.1038/s41562-018-0399-z.
- Jay Cao, Jacky Chen, John Hull, and Zissis Poulos. Deep hedging of derivatives using reinforcement learning. *arXiv preprint arXiv:2103.16409*, 2021. URL <https://arxiv.org/abs/2103.16409>.
- Jack Carr. *Applications of Centre Manifold Theory*. Springer, 1981.
- Marius Chevallier, Stefano De Marco, and Pierre-Emmanuel Lévy-dit Vehel. An optimal transport approach to arbitrage correction: Application to volatility stress-tests. *arXiv preprint arXiv:2501.12195*, 2025. URL <https://arxiv.org/abs/2501.12195>.
- Carl Chiarella, Xue-Zhong He, and Cars H. Hommes. The dynamics of moving average rules in a continuous-time financial market model. Technical Report RP 268, UTS Quantitative Finance Research Centre, 2006. URL <https://www.uts.edu.au/globalassets/sites/default/files/qfr-archive-03/QFR-rp268.pdf>.
- Vedant Choudhary, Sebastian Jaimungal, and Maxime Bergeron. FuNVol: Multi-asset implied volatility market simulator using functional principal components and neural SDEs. *Quantitative Finance*, 2024. doi: 10.1080/14697688.2024.2396977.
- Samuel N. Cohen, Christoph Reisinger, and Sheng Wang. Arbitrage-free neural-SDE market models. *Applied Mathematical Finance*, 2023.
- Haoying Dai. Beta-dependent gamma feedback and endogenous volatility amplification in option markets. *arXiv preprint arXiv:2511.22766*, 2025. URL <https://arxiv.org/abs/2511.22766>.
- Chukwuma Dim, Bjørn Eraker, and Grigory Vilkov. 0DTEs: Trading, gamma risk, and volatility propagation. *SSRN Working Paper*, 2024.
- Maximilian Engel, Jeroen S. W. Lamb, and Martin Rasmussen. Lyapunov exponents and shear-induced chaos for a Hopf bifurcation with additive noise. *Probability Theory and Related Fields*, 188, 2024. doi: 10.1007/s00440-024-01301-4.
- Robert Faff. PBFJ editorial ...engaging with responsible science. “OPEN FOR BUSINESS” — launching the PBFJ pre-registration publication initiative. *Pacific-Basin Finance Journal*, 2022. doi: 10.1016/j.pacfin.2022.101859.

- Adam Farago and Erik Hjalmarsson. Long-horizon stock returns are positively skewed. *Review of Finance*, 27(2):495–538, 2023. doi: 10.1093/rof/rfac021.
- William Feller. Two singular diffusion problems. *Annals of Mathematics*, 54(1):173–182, 1951.
- Vladimir Filimonov and Didier Sornette. Quantifying reflexivity in financial markets: Toward a prediction of flash crashes. *Physical Review E*, 85:056108, 2012. doi: 10.1103/PhysRevE.85.056108.
- Rüdiger Frey and Alexander Stremme. Market volatility and feedback effects from dynamic hedging. *Mathematical Finance*, 7(4):351–374, 1997. doi: 10.1111/1467-9965.00036.
- Nicolae Gârleanu, Lasse Heje Pedersen, and Allen M. Poteshman. Demand-based option pricing. *Review of Financial Studies*, 22(10):4259–4299, 2009. doi: 10.1093/rfs/hhp005.
- Jim Gatheral and Antoine Jacquier. Arbitrage-free SVI volatility surfaces. *Quantitative Finance*, 14(1):59–71, 2014. URL <https://arxiv.org/abs/1204.0646>.
- Igor Halperin and Andrey Itkin. Marketron through the looking glass: From equity dynamics to option pricing in incomplete markets. *arXiv preprint arXiv:2508.09863*, 2025. URL <https://arxiv.org/abs/2508.09863>.
- Stephen J. Hardiman, Nicolas Bercot, and Jean-Philippe Bouchaud. Critical reflexivity in financial markets: A Hawkes process analysis. *European Physical Journal B*, 86:442, 2013. URL <https://arxiv.org/abs/1302.1405>.
- John A. Hartigan and P. M. Hartigan. The dip test of unimodality. *Annals of Statistics*, 13(1):70–84, 1985. doi: 10.1214/aos/1176346577.
- Guangyi He, Tobias Sutter, and Lukas Gonon. Distributional adversarial attacks and training in deep hedging. In *Advances in Neural Information Processing Systems (NeurIPS)*, 2025. URL <https://arxiv.org/abs/2508.14757>.
- Xue-Zhong He, Kai Li, Junjie Wei, and Min Zheng. Market stability switches in a continuous-time financial market with heterogeneous beliefs. *Economic Modelling*, 26(6):1432–1442, 2009. doi: 10.1016/j.econmod.2009.07.016.
- Steven L. Heston. A closed-form solution for options with stochastic volatility with applications to bond and currency options. *Review of Financial Studies*, 6(2):327–343, 1993. doi: 10.1093/rfs/6.2.327.
- Bamdad Hosseini, Alexander W. Hsu, and Amirhossein Taghvaei. Conditional optimal transport on function spaces. *arXiv preprint arXiv:2311.05672*, 2023. URL <https://arxiv.org/abs/2311.05672>.
- Shengyi Huang et al. Open RL benchmark: Comprehensive tracked experiments for reinforcement learning. *arXiv preprint arXiv:2402.03046*, 2024. URL <https://arxiv.org/abs/2402.03046>.
- Zacharia Issa, Blanka Horvath, Maud Lemercier, and Cristopher Salvi. Non-adversarial training of neural SDEs with signature kernel scores. In *Advances in Neural Information Processing Systems (NeurIPS)*, 2023. URL <https://arxiv.org/abs/2305.16274>.
- Chen Jin and Ankush Agarwal. Forecasting implied volatility surface with generative diffusion models. *arXiv preprint arXiv:2511.07571*, 2025. URL <https://arxiv.org/abs/2511.07571>.

- Rafail Z. Khasminskii. *Stochastic Stability of Differential Equations*. Sijthoff and Noordhoff, 1980.
- Yuri A. Kuznetsov. *Elements of Applied Bifurcation Theory*. Springer, 3rd edition, 2004.
- Roger W. Lee. The moment formula for implied volatility at extreme strikes. *Mathematical Finance*, 14(3):469–480, 2004. doi: 10.1111/j.0960-1627.2004.00200.x.
- Giulia Livieri, Saad Mouti, Andrea Pallavicini, and Mathieu Rosenbaum. Rough volatility: Evidence from option prices. *IISE Transactions*, 50(9):767–776, 2018.
- Roger Lord, Remmert Koekkoek, and Dick van Dijk. A comparison of biased simulation schemes for stochastic volatility models. *Quantitative Finance*, 10(2):177–194, 2010. doi: 10.1080/14697680802392496.
- Qinmeng Luan and James Hamp. Automated regime classification in multidimensional time series data using sliced Wasserstein k -means clustering. *Data Science in Finance and Economics*, 2025. doi: 10.3934/DSFE.2025016.
- Shaocong Ma and Heng Huang. Robust reinforcement learning in finance: Modeling market impact with elliptic uncertainty sets. *arXiv preprint arXiv:2510.19950*, 2025. URL <https://arxiv.org/abs/2510.19950>.
- Tudor Manole, Sivaraman Balakrishnan, and Larry Wasserman. Minimax confidence intervals for the sliced Wasserstein distance. *Electronic Journal of Statistics*, 2022. URL <https://arxiv.org/abs/1909.07862>.
- Phillip Murray, Ben Wood, Hans Buehler, Magnus Wiese, and Mikko S. Pakkanen. Deep hedging: Continuous reinforcement learning for hedging of general portfolios across multiple risk aversions. *arXiv preprint arXiv:2207.07467*, 2022. URL <https://arxiv.org/abs/2207.07467>.
- Brian Ning, Sebastian Jaimungal, Xiaorong Zhang, and Maxime Bergeron. Arbitrage-free implied volatility surface generation with variational autoencoders. *SIAM Journal on Financial Mathematics*, 15(1), 2024. doi: 10.1137/21M1443546.
- Charles Packer, Katelyn Gao, Jernej Kos, Philipp Krähenbühl, Vladlen Koltun, and Dawn Song. Assessing generalization in deep reinforcement learning. *arXiv preprint arXiv:1810.12282*, 2018. URL <https://arxiv.org/abs/1810.12282>.
- Christophe Pérignon, Olivier Akmansoy, Christophe Hurlin, Anna Dreber, Felix Holzmeister, Jürgen Huber, Magnus Johannesson, Michael Kirchler, Albert J. Menkveld, Michael Razen, and Utz Weitzel. Computational reproducibility in finance: Evidence from 1,000 tests. *Review of Financial Studies*, 37(11):3558, 2024. doi: 10.1093/rfs/hhae029.
- Eckhard Platen and Martin Schweizer. On feedback effects from hedging derivatives. *Mathematical Finance*, 8(1):67–84, 1998. doi: 10.1111/1467-9965.00038.
- Dimitris N. Politis and Joseph P. Romano. The stationary bootstrap. *Journal of the American Statistical Association*, 89(428):1303–1313, 1994. doi: 10.1080/01621459.1994.10476870.
- Michael Roper. Arbitrage-free implied volatility surfaces. Technical report, University of Sydney, 2010.
- Thomas Schreiber and Andreas Schmitz. Improved surrogate data for nonlinearity tests. *Physical Review Letters*, 77(4):635–638, 1996. doi: 10.1103/PhysRevLett.77.635.

- K. Ronnie Sircar and George Papanicolaou. General Black–Scholes models accounting for increased market volatility from hedging strategies. *Applied Mathematical Finance*, 5(1):45–82, 1998.
- George Soros. The general theory of reflexivity. Open Society Foundations Working Paper, 2009.
- SqueezeMetrics. The implied order book and gamma exposure. Whitepaper, squeezemetrics.com, 2017.
- Adarsh Subbaswamy, Roy Adams, and Suchi Saria. Evaluating robustness to dataset shift via parametric robustness sets. In *Advances in Neural Information Processing Systems (NeurIPS)*, 2022. URL https://proceedings.neurips.cc/paper_files/paper/2022/file/6b7f9d9c1217a748391800871ff7d17d-Paper-Conference.pdf.
- Milena Vuletić and Rama Cont. VolGAN: A generative model for arbitrage-free implied volatility surfaces. *Applied Mathematical Finance*, 2025. doi: 10.1080/1350486X.2025.2471317.

Hawkes-SV equivalence at the Hopf boundary ($n_{SV}(\kappa^*) = 1$ by construction)

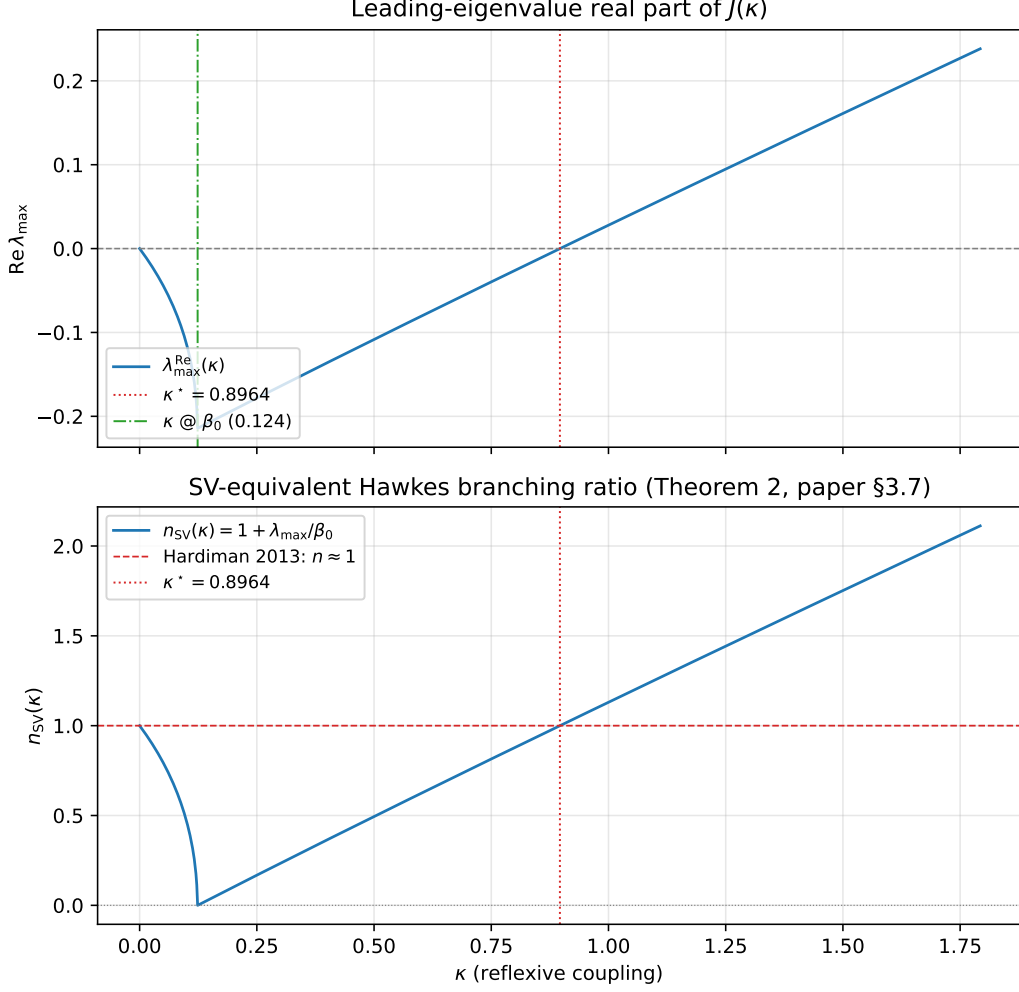


Figure 8: **SV-equivalent Hawkes branching ratio $n_{SV}(\kappa)$ at the canonical regime.** *Top:* leading-eigenvalue real part of $J(\kappa)$, with the gauge zero at $\kappa = 0$, the most-stable point at $\kappa_{\text{ref}} = 0.124$ (the node-spiral transition, $\beta_0 \approx 0.214$), and the Hopf threshold $\kappa^* = 0.8964$ where $\lambda_{\max} = 0$. *Bottom:* $n_{SV}(\kappa) = 1 + \lambda_{\max}/\beta_0$, anchored by $n_{SV}(\kappa_4^*) = 0.99996 \approx 1$ where the residual 4×10^{-5} is the truncation error in the 4-decimal $\kappa_4^* = 0.8964$; at machine-precision κ^* the identity is exact (Theorem 2, claim 1); the empirical Hardiman et al. (2013) $n \approx 1$ is the dashed red line. Past κ^* , $n_{SV} > 1$ (Hawkes-non-stationary regime). Run dir: `runs/hawkes_sv_equivalence/`.

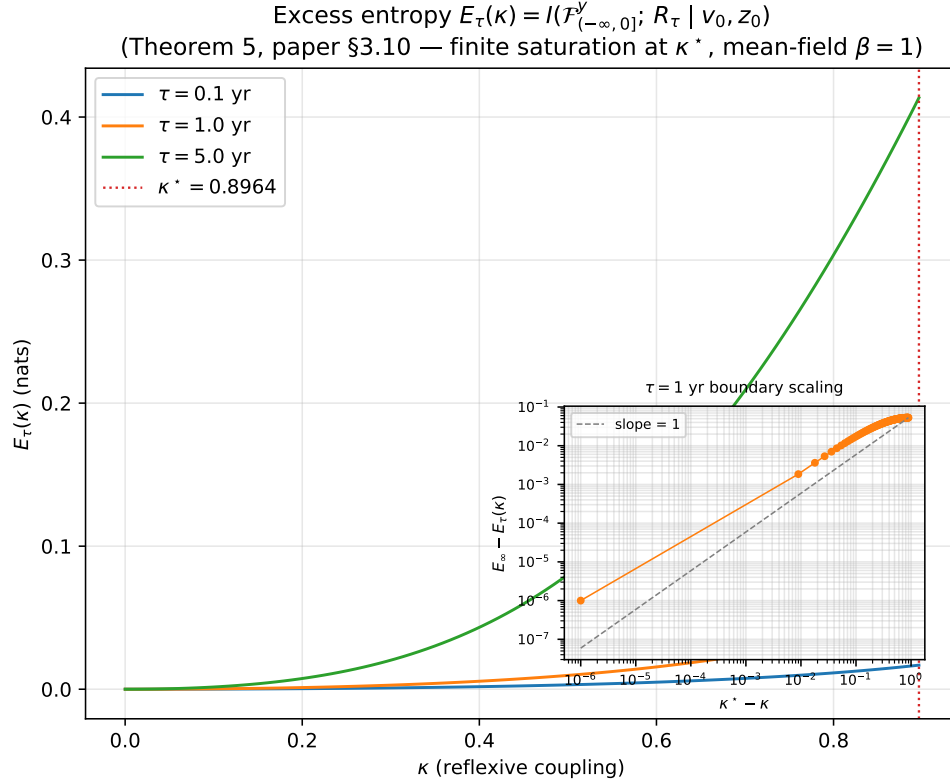


Figure 9: **Excess entropy** $E_\tau(\kappa)$ **at the canonical regime** for $\tau \in \{0.1, 1, 5\}$ yr. Markov closure $E_\tau(0) = 0$ at $\kappa = 0$; finite saturation $E_\tau(\kappa^*)$ at the Hopf threshold (dashed vertical); approach is linear in $(\kappa^* - \kappa)$ with mean-field critical exponent $\beta = 1$. Run dir: `runs/info_theoretic_excess_entropy/`.

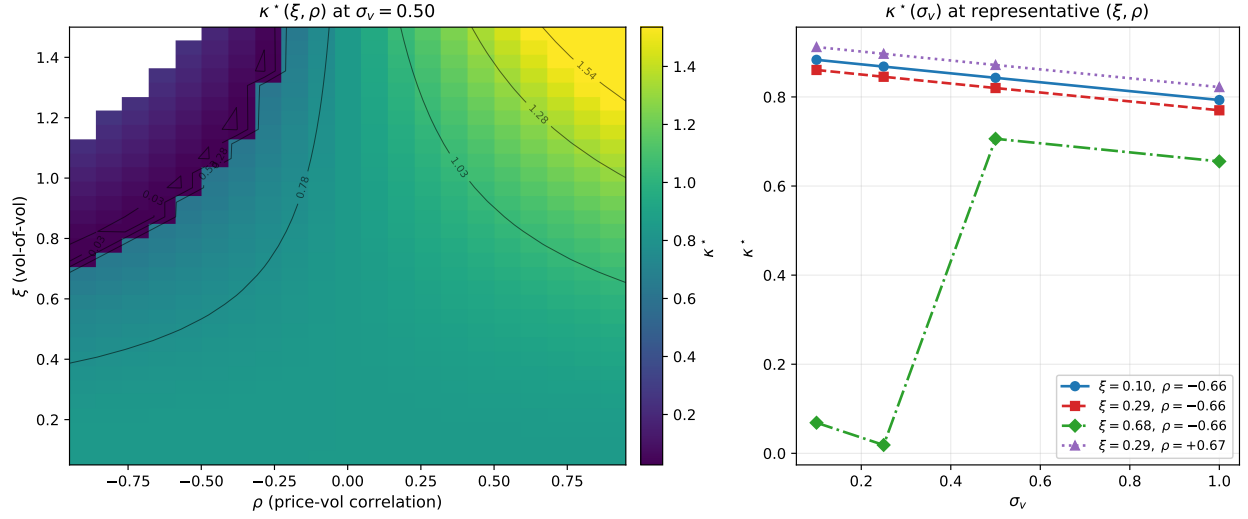


Figure 10: **Hopf bifurcation threshold κ^* across the (ξ, ρ, σ_v) slice of the reflexive parameter space.** *Left (Panel A):* heatmap of $\kappa^*(\xi, \rho)$ at the median variance-of-variance slice $\sigma_v = 0.50$, computed at the canonical Hopf-exhibiting regime of section 3.4 ($G_x = 0.5$, $G_v = -0.5$, $G_z = -0.5$, $\alpha = 0.5$, $\beta = 1$, $\gamma = 0.5$, $\kappa_v = 2$). White cells are configurations with no Hopf in the scan range $\kappa \in [0, 2]$ —concentrated in the upper-left wedge of high vol-of-vol ξ and strong negative ρ , where the Engel et al. (2024) shear correction destabilises the eigenvalue envelope. Contour lines are isoquants of κ^* at increments of $(\kappa_{\max}^* - \kappa_{\min}^*)/6$. *Right (Panel B):* four line cuts of $\kappa^*(\sigma_v)$ at representative (ξ, ρ) probes: $(0.10, -0.70)$, $(0.30, -0.70)$, $(0.70, -0.70)$, and $(0.30, +0.70)$. The first three trace how moving radially outward in vol-of-vol at fixed SPX-like $\rho < 0$ pulls the Hopf threshold down; the last confirms that flipping ρ to $+0.70$ at moderate ξ keeps κ^* in the 0.85–0.95 band.

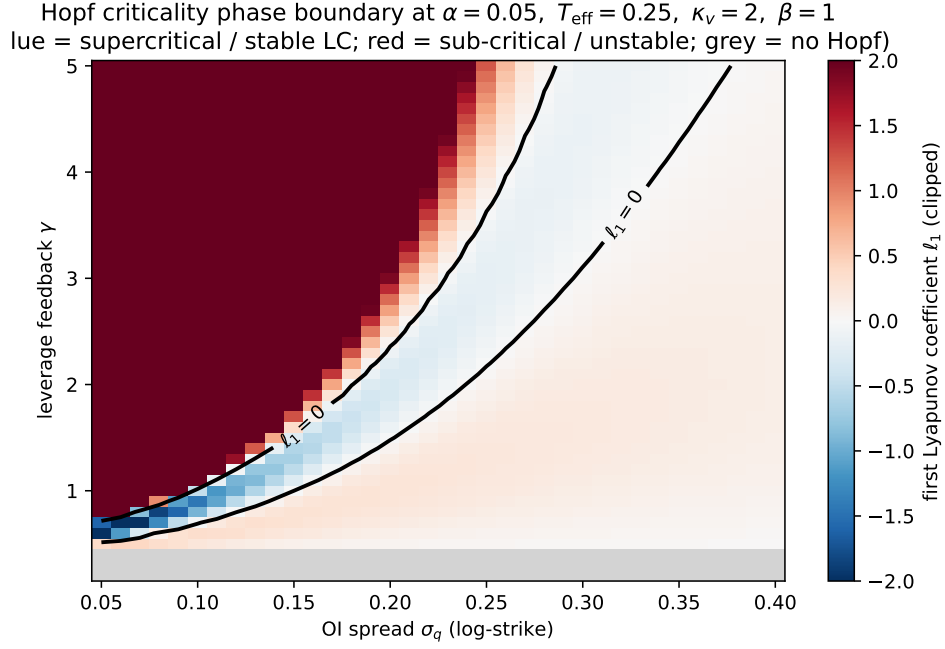


Figure 11: **Closed-form ℓ_1 phase boundary in (σ_q, γ) space at the canonical specification.** Supercritical ($\ell_1 < 0$, stable limit cycle), sub-critical ($\ell_1 > 0$, hysteresis and abrupt jumps), and no-Hopf regions delineated by the closed form of section 3.5. Computed in $\sim 50 \mu\text{s}$ per cell.

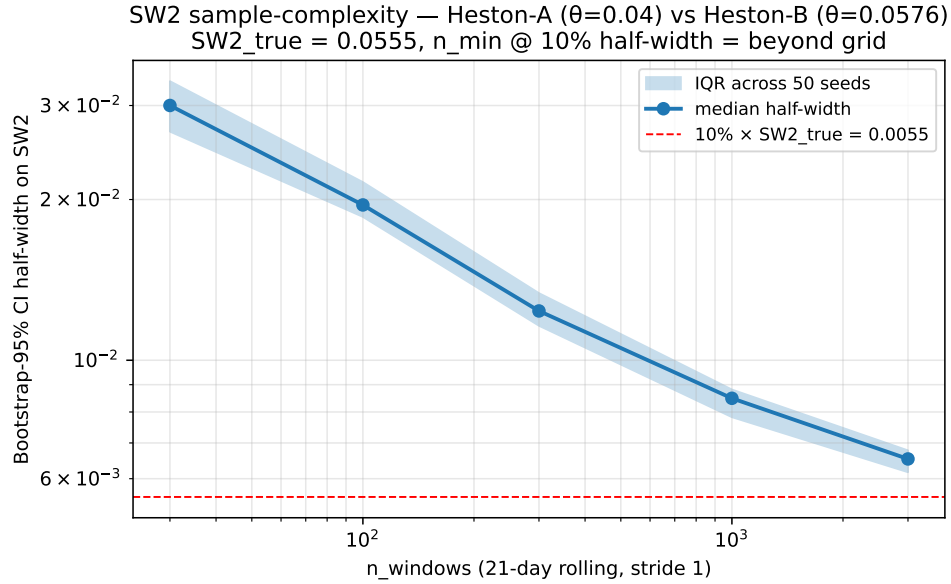


Figure 12: **Sliced-W2 bootstrap-95%-CI half-width vs n_{windows} (log-log).** Shaded band = IQR across 30 outer seeds; dashed red line = $10\% \cdot \text{SW2}_{\text{true}}$ target.

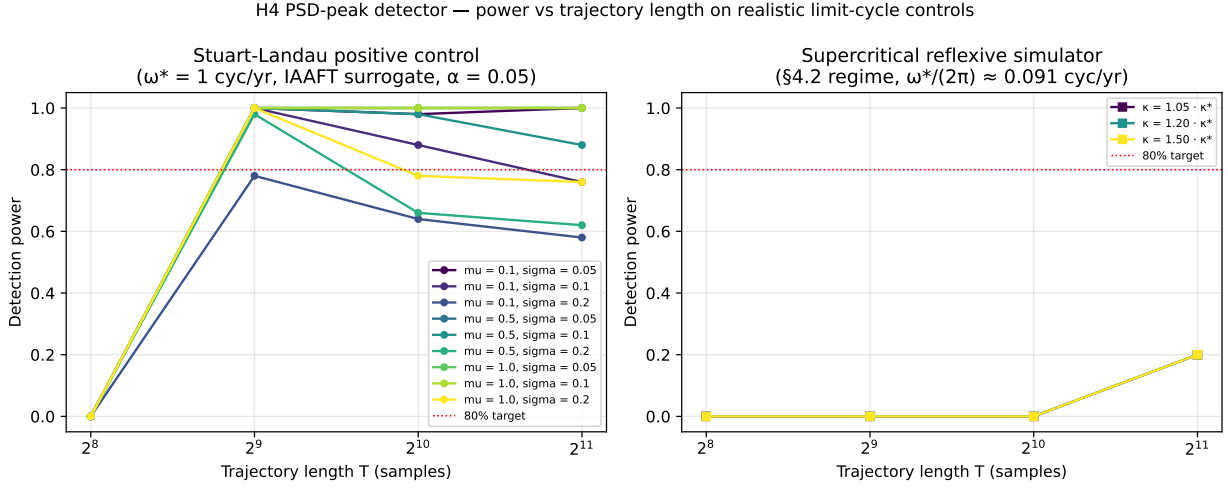


Figure 13: **H4 detector power against realistic limit-cycle positive controls.** *Left:* Stuart–Landau real-part signal at $\omega^* = 1$ cyc/yr; 8/9 (μ, σ) configurations attain $\geq 80\%$ power at $T_{\min} = 512$ (the 9th — $\mu = 0.1, \sigma = 0.20$, the weakest cycle and noisiest regime — saturates at $\sim 78\%$). *Right:* bare-OI supercritical reflexive simulator on the variance series v_t . Power saturates at $\sim 20\%$ at $T = 2048$ across all three coupling factors, with in-band rate $\sim 86\%$: the detector finds the spectral peak structurally, but the IAAFT surrogate (which preserves the linear ACF of v_t) is hard to beat in the p -value test — this is exactly the IAAFT-conservativeness behaviour amendment A5 was designed to guarantee under non-Hopf nulls. Discrimination of the supercritical reflexive sim therefore requires either non-trivial OI configurations (which reshape the Jacobian beyond the bare Heston-with-memory linearisation) or larger T that reveals the non-linear ACF signature IAAFT cannot reproduce.

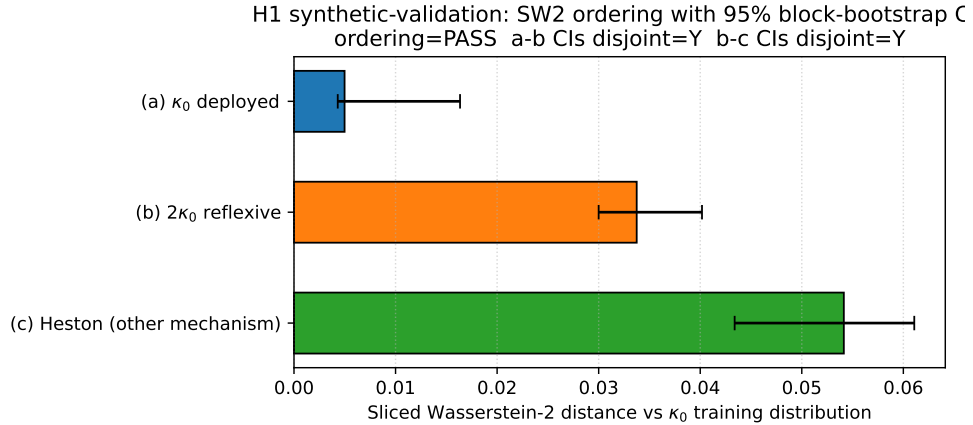


Figure 14: **H1 synthetic-validation ordering: SW2 vs the κ_0 -trained reference distribution with 95% block-bootstrap CIs.** The locked H1 protocol returns the expected ordering $\text{SW2}(\kappa_0\text{-deployed}) < \text{SW2}(2\kappa_0\text{-reflexive}) < \text{SW2}(\text{Heston})$ with non-overlapping CIs at the production-budget run. This converts the H1 protocol from “designed but unrun” to “demonstrated working on synthetic ground truth — only the empirical SPX target is missing.”

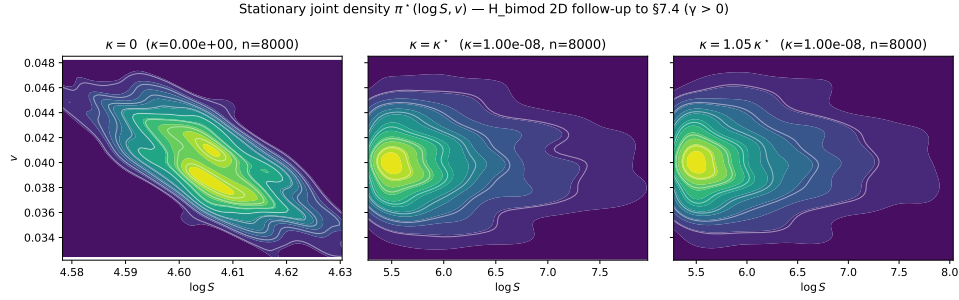


Figure 15: **Joint $(\log S, v)$ stationary density across the κ sweep.** 2D KDE contours at $\kappa \in \{0, \kappa_{\text{env}}^*, 1.05 \kappa_{\text{env}}^*\}$ with $\gamma > 0$ active, where $\kappa_{\text{env}}^* \approx 3.9 \times 10^{-9}$ is the simulator’s *stability-envelope* upper bound (*not* the Hopf threshold $\kappa^* \approx 0.896$ of section 3.4 — the two are distinct objects defined at different parameter scales). The PCA principal direction at $\kappa = 1.05 \kappa_{\text{env}}^*$ has explained-variance ratio 0.564 and a Hartigan dip $p = 0.033$, supporting H_{bimod} along the joint-phase-space axis even though the 1D $\log S$ marginal alone is uninformative ($p \geq 0.10$ across all κ). Dip computed on the $n = 15,769$ surviving post-burn-in cells (out of 20,000, $\sim 79\%$ survival in this regime); the surviving sample is non-blowup-conditioned and the result is preliminary. Run dir: `runs/h_bimod_2d/`.